

Distance-Shell Tomography: Geometric Certificates for Fault-Tolerant Sensing, Moment Compression, and Hypercube Equalization

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Abstract

Distance-shell tomography observes an anonymous integer population on a graph through the multiset of its distances to each labelled sensor. Its ambiguities are signed integer trades, and every sensor either detects a trade or is blind to it. We develop this detector calculus into exact theorems for three questions: how much redundancy survives lost reports, how far reports can be compressed, and how few sensors suffice to equalize every outside pair. On the $n \times n$ grid, recovery of all populations of mass at most two after one sensor erasure is, once the four corners are instrumented, equivalent to a family of interval cuts, and the minimum number of sensors is exactly $\lceil 3(n+1)/2 \rceil$ for every $n \geq 3$; without the corner requirement we prove the lower bound $\lceil (3n-4)/2 \rceil$, one larger when $n \equiv 2 \pmod{4}$, and exhibit placements attaining it for every $5 \leq n \leq 20$. All but four monitors may report a single distance sum. On a d -dimensional Hamming product with $d \geq 1$ with at least two values per coordinate, a single k th distance moment has population capacity exactly $2^k - 1$ for $k < d$ and unrestricted capacity for $k \geq d$. On Q_d , its report distance is exactly 2^{d-k} when $d \geq 2k \geq 2$ and the population bound satisfies $2^{k-1} \leq h < 2^k$. For hypercubes of dimension divisible by four we prove $2^{n-1} + n/2 - 1 \leq \xi(Q_n) \leq 2^{n-1} + n/2$ for the equidistant dimension, refuting a conjectured exponential correction and leaving an additive gap of one; the upper value is exact for $n = 8, 12, 16$. The same certificate method proves the even bishop-board conjecture $\gamma_2(B_n) = 2n$ and contradicts three further sourced conjectured statements. Complete proofs, a Lean development whose principal endpoints pass three independent kernels, and reproducible exact computations accompany the results.

1 Introduction

Several anonymous objects are harder to localize than one object, even when every sensor reports exact distances. The missing information is the correspondence between objects seen at different sensors. *Distance-shell tomography* [30] retains the multiplicity of each reported distance and the identity of each sensor, and discards the identities of the targets. A population is then an integer array, a sensor report is a linear observation of it, and two populations are confused exactly when their difference is a signed integer *trade* that the selected sensors cannot see.

This paper turns that viewpoint into a calculus of *detectors*. Every signed population z splits the sensors into those that detect it and those that are blind to it. Three classical-looking questions become statements about how a sensor set meets these two families.

- **Redundancy.** A placement survives e lost reports exactly when it meets every detector set at least $e + 1$ times. How many sensors does that cost?
- **Compression.** A sensor may report a single moment of its distance distribution. Which detectors survive, and how many populations can still be told apart?

- **Blindness.** A distance-equalizer set must contain, for every outside pair, a sensor blind to their dipole. How small can it be?

In each case the answer is obtained by isolating a small family of exceptional trades, describing their detector or blind sets geometrically, and certifying an extremal design against that family. We call the resulting short, checkable objects *geometric certificates*: interval cuts, boundary incidences, parity characters, tensor trades, and modular weights.

1.1 Main results

Let $G_n = P_n \square P_n$ be the square grid on $\{0, \dots, n-1\}^2$. Let $R_2(n)$ be the smallest number of sensors that resolve all populations of mass at most two after any one sensor erasure, and $R_2^\square(n)$ the same minimum over placements containing the four corners. The predecessor [30] proved that $n-2$ sensors are optimal without erasures for $n \geq 6$ and asked for the cost of redundancy.

Theorem 1 (Grid redundancy). *For every $n \geq 3$,*

$$R_2^\square(n) = \left\lceil \frac{3(n+1)}{2} \right\rceil.$$

Without the corner requirement,

$$\left\lceil \frac{3n-4}{2} \right\rceil + \varepsilon_n \leq R_2(n) \leq \left\lceil \frac{3(n+1)}{2} \right\rceil, \quad \varepsilon_n = \begin{cases} 1, & n \equiv 2 \pmod{4} \text{ and } n \geq 6, \\ 0, & \text{otherwise,} \end{cases}$$

and the lower bound is attained for every $5 \leq n \leq 20$. In particular $R_2(n) = 3n/2 + O(1)$, the two bounds differ by at most four, and instrumenting all four corners costs exactly three or four sensors in the verified range.

The additive deletion bound for one erasure is $n-1$; the theorem shows that half as many sensors again are necessary and sufficient. The proof rests on an exact reduction (Theorem 8): with corners present, every unresolved pair of two-target populations forces a detector set containing the symmetric difference of a row interval and a column interval of equal length, and every such *interval detector* is realized by an explicit trade. The reduction also shows that all but four monitors can compress their report to one integer (Corollary 9), and it extends to rectangles, where long boards have optimum $2N+4$ and rigid optimal designs (Theorem 18).

Theorem 2 (Hypercube blindness). *Let n be a positive multiple of four. The equidistant dimension of the n -cube satisfies*

$$2^{n-1} + \frac{n}{2} - 1 \leq \xi(Q_n) \leq 2^{n-1} + \frac{n}{2},$$

with $\xi(Q_4) = 9$, $\xi(Q_8) = 132$, $\xi(Q_{12}) = 2054$, and $\xi(Q_{16}) = 32776$.

This is Theorem 26. It refutes the conjecture $\xi(Q_n) = 2^{n-1} + 2^{n/2-2}$ of [13] for every $n \geq 12$ divisible by four and, with the known value 2^{n-1} in the other dimensions, determines the equidistant dimension of every hypercube up to an additive error of one. The upper bound is a parity-restricted prefix-balancing construction. The lower bound comes from a quantitative form of the balancing theorem of Alon, Bergmann, Coppersmith, and Odlyzko (Theorem 34): the product of the first-moment fields of antipodal dipoles is a low-degree polynomial on the sign cube, and too few factors leave a set of unbalanced points on which no low-degree polynomial can vanish.

Theorem 3 (Moment compression). *On a Hamming product of $d \geq 1$ complete graphs, each with at least two vertices, the field of a single k th distance moment determines exactly the marginals on at most $\min(k, d)$ coordinates; for $k < d$ it resolves all populations of mass at most $2^k - 1$ and no more. On Q_d with $d \geq 2k \geq 2$, the reports of distinct populations of mass at most h , $2^{k-1} \leq h < 2^k$, differ at no fewer than 2^{d-k} sensors, and this is attained.*

These are Theorems 21 and 23. They are sharp sensing corollaries of classical Hamming-scheme, null-design, and eigenfunction-support theory, recorded here with their exact observation contracts.

Finally, the certificate method reaches domination problems outside tomography. Boundary parity proves the conjecture $\gamma_2(B_n) = 2n$ for even bishop boards (Theorem 41); modular zero-sum weights give $\gamma_{\times 2,t}(P(11,4)) = 16$ and $\gamma_{\times 2,t}(P(16,5)) = 24$, contradicting two conjectured formulas (Theorem 42); and a six-vertex graph refutes a conjecture on fractional domination and packing faces (Theorem 45). In total the paper contradicts four sourced conjectured statements and proves a fifth.

1.2 Antecedents and attribution

The predecessor supplies the observation model, integer trades, the optimal erasure-free grid construction, and the redundancy question [30]. Multiple-access channels already record coordinatewise symbol multiplicities [9]. Resolving sets and metric dimension go back to [31, 17]; fault-tolerant resolving sets to [19]; error correction from graph distances for single-vertex messages to [4]. Several-object metric dimension with nearest-object reports uses a different information contract [25].

The singleton interval constraints of the grid problem are double total domination in a rook graph, with additional boundary incidences. The $3/2$ coefficient and unions of three-leaf stars are already present in that theory [21, 7], as are the elongated $5/3$ regime and the two-column design used for rectangles; we do not claim them as new there. The contribution here is the exact correspondence with population ambiguities, the all-interval requirement, the boundary accounting that makes the corner-constrained optimum exact for every n , and the resulting tomography theorems.

For hypercubes, full-parity containment and the odd-bisector argument are known [14, 13, 24]; the antipodal quotient is an existing graph [5]; prefix balancing and the parity lemma are classical [22, 2]. Our candidate contributions are the exact reduction to ordinary domination of the folded graph, the quantitative balancing theorem and the resulting bound within one, and the exact values. The moment section combines established spectral, null-design, interpolation, and switching constructions [8, 26, 11, 33]; it is presented as a collection of sensing corollaries. Failure to locate an earlier exact statement is not a proof of priority, and mathematical validity, formal coverage, and first-discovery priority are kept distinct throughout.

1.3 Organization

Section 2 sets up the calculus. Sections 3 and 4 prove Theorem 1; Section 5 treats rectangles and additive fields; Section 6 treats moments; Section 7 treats hypercubes. Sections 8, 9, and 10 contain the domination certificates. The final sections discuss the distributed interpretation, the verification record, and open problems.

2 The distance-shell calculus

Let G be a finite connected graph with distance dist . A *population* is a function $x : V(G) \rightarrow \mathbb{N}$, of mass $|x| = \sum_v x(v)$; the empty population and repeated vertices are allowed. At a sensor $s \in V(G)$ the *shell report* is the multiset containing $x(v)$ copies of $\text{dist}(s, v)$ for every v . Equivalently, its radius- r entry is

$$H_s x(r) = \sum_{v: \text{dist}(s,v)=r} x(v).$$

The graph is known, sensors are distinct vertices, and sensor labels are retained. The same formula defines $H_s z$ for a signed population $z : V(G) \rightarrow \mathbb{Z}$.

Definition 4. A sensor s detects a signed population z if $H_s z \neq 0$ and is blind to it otherwise. The detector set is $D(z) = \{s : H_s z \neq 0\}$, and $D(x, y) = D(x - y)$ for two populations. A nonzero z is a trade for a sensor set S if $S \cap D(z) = \emptyset$.

Two populations have equal reports on S exactly when their difference is a trade for S . Faults are governed by the same sets.

Proposition 5 (Detector weight is fault tolerance). *Let $\mathcal{X}$ be a set of populations, S a set of labelled sensors, and e, t nonnegative integers. Every member of $\mathcal{X}$ is determined by the reports on S after at most e identified erasures if and only if $|S \cap D(x, y)| \geq e+1$ for all distinct $x, y \in \mathcal{X}$. It is determined after at most t arbitrarily corrupted reports if and only if $|S \cap D(x, y)| \geq 2t+1$.*

Proof. The word $(H_s x)_{s \in S}$ has one block per sensor, and the block Hamming distance between the words of x and y is $|S \cap D(x, y)|$. An erasure pattern confuses x and y exactly when it covers every differing block. Two words at distance at most $2t$ have a common word within distance t of both, and conversely. This standard Hamming criterion concerns whole sensor reports; it makes no claim about numerical perturbations within a report. $\square$

Shell polynomials and moments. The report of z at s is encoded by the polynomial

$$R_s(z; T) = \sum_v z(v) T^{\text{dist}(s, v)} = \sum_{r \geq 0} H_s z(r) T^r, \quad (1)$$

which is nonzero exactly when s detects z . Its moments

$$M_k z(s) = \sum_v z(v) \text{dist}(s, v)^k = \left(T \frac{d}{dT} \right)^k R_s(z; T) \Big|_{T=1}$$

are scalar compressions of the report: M_0 is the mass, M_1 the total distance. Distances add on Cartesian products, so shell polynomials of tensor trades multiply and their detector sets are products (Theorem 24).

Dipoles: resolving and equalizing. For the dipole $\delta_x - \delta_y$ of two distinct vertices,

$$s \notin D(\delta_x - \delta_y) \iff \text{dist}(s, x) = \text{dist}(s, y). \quad (2)$$

A resolving set meets every dipole detector set; a fault-tolerant resolving set meets each twice. A distance-equalizer set meets the complementary *blind set*, the bisector, of every dipole supported outside it. Metric dimension and equidistant dimension are thus the two mass-one extremes of the calculus. Sections 3–6 study detectors of populations of mass two and more; Section 7 studies blind sets.

3 Square grids: two-target trades and interval cuts

Let $G_n = P_n \square P_n$ and put $m = n - 2$. By Proposition 5, a placement S survives one erasure for populations of mass at most two precisely when

$$|S \cap D(x, y)| \geq 2 \quad \text{for all distinct } x, y \text{ with } |x|, |y| \leq 2. \quad (3)$$

Different masses differ at every sensor. Opposite corners have equivalent reports at a fixed mass, since their distances to a vertex sum to $2(n - 1)$. Thus a population pair detected at a corner is also detected at the opposite corner. Two adjacent corners distinguish a single vertex. With four corners present, the remaining obstruction in (3) is therefore a distinct pair of mass-two populations agreeing at all corners.

Write $[m] = \{1, \dots, m\}$ for the interior coordinate set. If $I, J \subseteq [m]$ are nonempty intervals of equal length, define the *interval detector*

$$D(I, J) = \{(x, y) : (x \in I) \text{ xor } (y \in J)\}. \quad (4)$$

The symmetric difference in (4) is essential: the intersection rectangle is excluded.

Lemma 6 (Exact skinny trades). *Let $I = \{a + 1, \dots, a + k\}$ and $J = \{b + 1, \dots, b + k\}$ be contained in $[m]$, with $k \geq 1$. The populations*

$$X = \{(a, b + 1), (a + k + 1, b + k)\}, \quad Y = \{(a + 1, b), (a + k, b + k + 1)\}$$

are distinct and have detector set exactly $D(I, J)$.

Proof. Both populations lie in the grid. Their two-element multisets of coordinate sums agree, as do their multisets of coordinate differences. If a sensor lies outside both intervals, the appropriate affine formula for Manhattan distance gives equal bags. If it lies inside both, set $u = x - a$, $v = y - b$, so $1 \leq u, v \leq k$. Both bags are

$$\{u + v - 1, 2k + 1 - u - v\}.$$

If exactly one coordinate is inside, the sums of the two reported distances differ by 2 or -2 : the interior coordinate contributes a nonzero difference, while the exterior coordinate contributes zero. The reports therefore differ. These cases also include sensors on the bounding lines, which are outside the open intervals. Finally $(a, b + 1)$ is in X and is in neither position of Y . $\square$

Lemma 7 (Bounding intervals of a corner collision). *If distinct mass-two populations X, Y agree at all four corners, there are nonempty equal-length intervals $I, J \subseteq [m]$ such that $D(I, J) \subseteq D(X, Y)$.*

Proof. Use diagonal coordinates $u = x + y$, $v = x - y$. Corner equality gives identical multisets of both coordinates. Distinct two-element populations must use the two different matchings of two distinct u -values and two distinct v -values. The four intersections are lattice points, so the differences in each diagonal coordinate are positive even integers.

Up to interchanging populations and reflecting a grid coordinate, their positions have the form

$$\begin{aligned} X &= \{(a, b + t), (a + s + t, b + s)\}, \\ Y &= \{(a + t, b), (a + s, b + s + t)\}, \quad s, t \geq 1. \end{aligned}$$

Both coordinate bounding intervals have length $L = s + t$. Let I, J be their open integer interiors. They are nonempty, have length $L - 1$, and lie inside $[m]$.

To see that every sensor in their symmetric difference detects the pair, take the common center c , put $\alpha = (s + t)/2$ and $\beta = |s - t|/2$, and write $p = |x - c_x|$, $q = |y - c_y|$. The difference between the sums of the two distances is

$$2[\max(p, \alpha) - \max(p, \beta) - \max(q, \alpha) + \max(q, \beta)].$$

If $x \in I$ and $y \notin J$, then $p < \alpha \leq q$. The first difference of maxima is positive and the second is zero. The other case has the opposite sign. Reflection preserves interval membership and detector inclusion. $\square$

Theorem 8 (Exact interval-cut criterion). *A placement containing all four corners survives any one erasure for all populations of mass at most two if and only if*

$$|S \cap D(I, J)| \geq 2$$

for every nonempty pair of interior intervals I, J of equal length. Necessity holds without the corners.

Proof. Necessity is Lemma 6. For sufficiency, different masses and pairs detected by corners already have two detectors. Distinct single-target populations are detected by corners. Every remaining pair has mass two and agrees at all corners, so Lemma 7 supplies the required interval detector subset. $\square$

This theorem describes the entire additional obstruction after corner sensing. Merely testing singleton intervals is insufficient: an ordered rectangle can contain a balanced union of several disconnected sensor components and have no crossing sensor at all.

Corollary 9 (Distance-sum compression). *Keep full multiset reports at the four corners, but at every other sensor s report only the total distance $M_1x(s)$. For populations of mass at most two and one erased report, a placement containing the corners is robust under these compressed reports if and only if it meets every equal-length interval detector at least twice. Consequently the corner-constrained optimum of Theorem 1 is unchanged under this reporting scheme.*

Proof. For necessity, a skinny trade has identical full reports outside its interval detector, and hence identical compressed reports there as well. For sufficiency, pairs distinguished by corners retain two full-report detectors. For every remaining pair, the proof of Lemma 7 distinguishes the sums of distances at every sensor of its interval detector. The interval condition supplies two such sensors. $\square$

Thus only four monitors need retain the distance distribution; all other monitors can compress their output to a single integer. This conclusion uses the full corner reports to handle different masses and initial geometric ambiguity. It does not assert that distance sums alone resolve every population.

4 Exact sensor budgets on square grids

4.1 A lower bound with boundary accounting

Associate one vertex to each interior row and interior column. An interior sensor becomes an edge of this bipartite incidence graph. A noncorner boundary sensor has one incident interior line and becomes a half-edge. Corners have no such incidence. Let $N = |S|$, let B count the half-edges, and let C count the corners.

For interior row i and column j , let r_i, c_j be their degrees, including half-edges, and let x_{ij} indicate an interior sensor at their intersection. The singleton case of Lemma 6 gives

$$r_i + c_j - 2x_{ij} \geq 2. \quad (5)$$

Proposition 10 (Boundary-sensitive charging bound). *Every placement satisfying (5) obeys*

$$4N \geq 6m + B + 4C. \quad (6)$$

If some interior line has degree zero, then moreover $N - C \geq 2m$. If every interior line has positive degree and equality holds in (6), the placement is ideal: every interior line has degree three or one; every line of degree three, a center, is joined by three interior sensors to three lines of degree one, its leaves; and every remaining line, a half-line, carries exactly one boundary sensor and no interior sensor.

Proof. First suppose every interior line has positive degree. Let L, M, H count line vertices of degrees 1, 2, and at least 3, respectively, and let h be the sum of degrees in the last class. A degree-one vertex whose unique incidence is an ordinary edge has a neighbor of degree at least three by (5). Charging such leaves to their distinct high-degree incidences, and the remaining leaves to their half-edges, gives $L \leq h + B$. Also $3H \leq h$.

The degree sum is $D = L + 2M + h$, with $D + B = 2(N - C)$ and $L + M + H = 2m$. Consequently

$$\begin{aligned} 4(N - C) - 6m &= 2(D + B) - 3(L + M + H) \\ &= M + (h + B - L) + (h - 3H) + B \geq B. \end{aligned}$$

This proves (6) in this case. Equality forces $M = 0$, $h = 3H$, and $L = h + B$. The last equation says that the charging is a bijection: every incidence at a line of degree three is an ordinary edge whose other end is a leaf, and every half-edge lies on a line of degree one. This is the ideal structure.

If some row has degree zero, every column has degree at least two by (5). Let E be the number of interior sensors and B_R, B_C the boundary sensors incident to rows and columns. The column degrees sum to $E + B_C \geq 2m$, so $N - C = E + B_R + B_C \geq 2m$, and $B_C \leq 2m$ since each interior column has two boundary vertices. Hence

$$4(N - C) = 4(E + B_R + B_C) \geq 8m + 4B_R \geq 6m + B_R + B_C.$$

The case of an empty column is symmetric. □

The full interval pair $I = J = [m]$ has exactly the noncorner boundary sensors as detectors. Therefore every robust placement has $B \geq 2$, and a robust placement containing the four corners has

$$4N \geq 6m + 18, \quad \text{that is,} \quad N \geq \left\lceil \frac{3m + 9}{2} \right\rceil = \left\lceil \frac{3(n + 1)}{2} \right\rceil. \quad (7)$$

4.2 Leaf-star designs

Proposition 10 tells us what an optimal design must look like: three-leaf stars and half-lines. An ideal placement with λ center rows, ν center columns, ρ half rows, and κ half columns has

$$\lambda + 3\nu + \rho = m = 3\lambda + \nu + \kappa, \quad B = \rho + \kappa, \quad N - C = 3\lambda + 3\nu + \rho + \kappa. \quad (8)$$

Write $m = 4q + r$ with $0 \leq r \leq 3$ and choose

$$(\lambda, \nu, \rho, \kappa) = \begin{cases} (q, q - 1, 3, 1), & r = 0, \ q \geq 1, \\ (q, q, 1, 1), & r = 1, \\ (q, q, 2, 2), & r = 2, \\ (q + 1, q, 2, 0), & r = 3. \end{cases} \quad (9)$$

The *leaf-star placement* $\mathcal{L}_m$ on the $(m + 2) \times (m + 2)$ grid consists of the four corners together with

$$\begin{aligned} (i, 3(i - 1) + t), & \quad 1 \leq i \leq \lambda, \ t \in \{1, 2, 3\}, & \text{(row-centered stars)} \\ (\lambda + i, 0), & \quad 1 \leq i \leq \rho, & \text{(half rows)} \\ (\lambda + \rho + 3(j - 1) + t, 3\lambda + \kappa + j), & \quad 1 \leq j \leq \nu, \ t \in \{1, 2, 3\}, & \text{(column-centered stars)} \\ (0, 3\lambda + j), & \quad 1 \leq j \leq \kappa. & \text{(half columns)} \end{aligned} \quad (10)$$

In the row order the centers come first, then the half rows, then the leaves of the column-centered stars. Figure 1 shows $\mathcal{L}_7$.

Lemma 11 (Leaf-star interval cuts). *For every $m \geq 1$, every nonempty pair of equal-length interior intervals has at least two detectors in $\mathcal{L}_m$.*

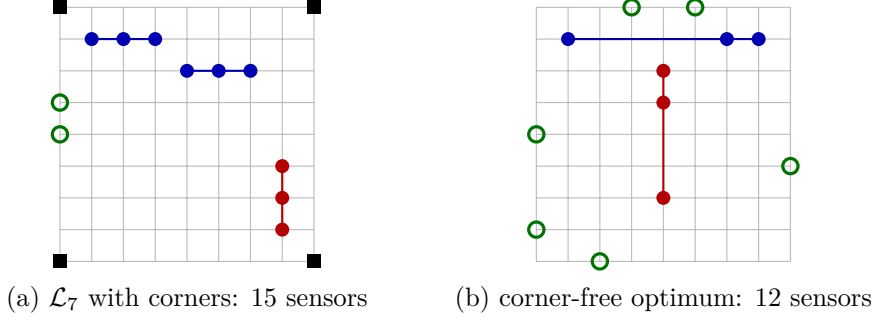


Figure 1: The 9×9 grid, with the first coordinate increasing downward. (a) The leaf-star placement: two row-centered stars, two half rows on the left boundary, one column-centered star, and the four corners. Every interior line has odd degree. (b) A corner-free placement attaining the lower bound of Theorem 14; it is again a union of three-leaf stars and half-lines, with six boundary sensors.

Proof. Let I and J be intervals of rows and columns with $|I| = |J| = k \geq 1$. A sensor is a detector when exactly one of its two lines is selected. Summing over the sensors, the number of detectors is

$$\sum_{i \in I} r_i + \sum_{j \in J} c_j - 2 \#\{\text{sensors with row in } I \text{ and column in } J\},$$

because corners and the boundary coordinate of a half-line sensor are never selected. Every interior line of $\mathcal{L}_m$ has degree three or one, so the count is congruent to $|I| + |J| = 2k$ modulo two. It suffices to exclude a zero cut.

Suppose no sensor is a detector. A selected half row would make its boundary sensor a detector, so I contains no half row; similarly J contains no half column. Since $\rho \geq 1$ and the half rows separate the centers from the leaf rows, the interval I lies among the center rows or among the leaf rows. In the first case each selected center forces its three leaf columns into J , and distinct centers have distinct leaves, so $|J| \geq 3|I| > |I|$. In the second case no center row is selected, so no leaf column of a row-centered star is selected; hence J consists of center columns, each of which forces its three leaf rows into I , so $|I| \geq 3|J|$ and $k \geq 3k$. Both contradict $|I| = |J| = k \geq 1$. $\square$

Proof of the corner-constrained formula in Theorem 1. The lower bound is (7). By Lemma 11 and Theorem 8, the placement $\mathcal{L}_m$ is robust. By (8) and (9) it has $6q + 5$, $6q + 6$, $6q + 8$, $6q + 9$ sensors for $r = 0, 1, 2, 3$, which is $\lceil (3m + 9)/2 \rceil$ in every case. $\square$

The parity step is the reason the design is so economical: with all line degrees odd, an interval cut can only be $0, 2, 4, \dots$, and the half rows placed at the junction of the row order make zero impossible. Equality in (6) holds for $\mathcal{L}_m$, with $B = 2$ when m is odd and $B = 4$ when m is even; the ideal equations (8) force $B \equiv 2m \pmod{4}$, which is exactly the rounding in (7).

4.3 The ordered-star variant

For $m = 4q$ there is a second optimal family without half-lines, obtained by ordering $2q$ stars and splitting one edge. It was the first construction found and is the one whose robustness was formalized first; we record it because its zero-cut analysis is different. Temporarily index interior lines by $0, \dots, 4q - 1$. Select

$$E_A = \{(i, 3i + t) : 0 \leq i < q, 0 \leq t < 3\}, \quad (11)$$

$$E_B = \{(q + 3(q - 1 - j) + t, 3q + j) : 0 \leq j < q, 0 \leq t < 3\}. \quad (12)$$

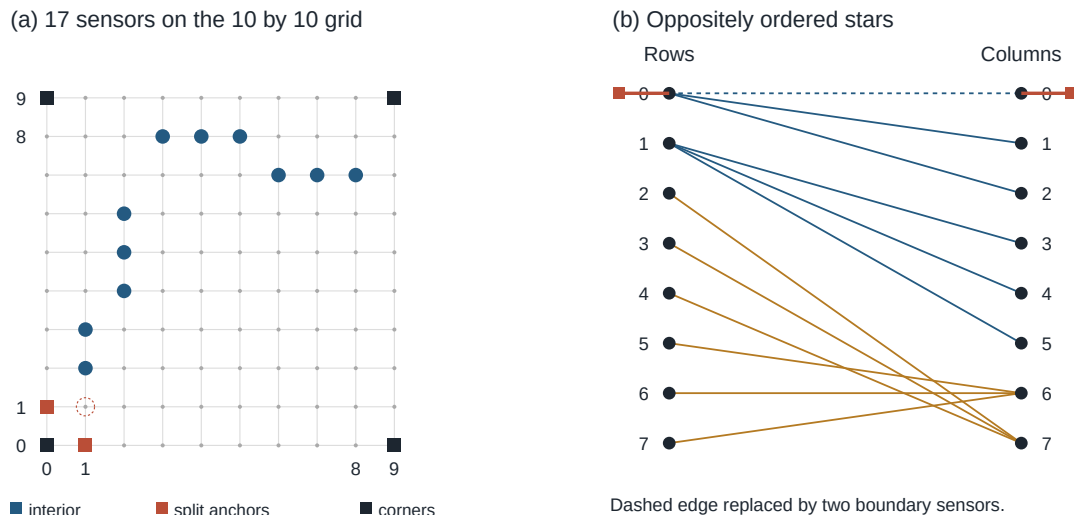


Figure 2: The ordered-star construction for $q = 2$. Interior line labels in panel (b) start at zero; grid coordinates in panel (a) are shifted by one. Splitting the dashed edge adds one sensor overall and supplies two detectors for the full interior interval pair.

There are $6q$ distinct edges. The reversal in (12) is the decisive ordering choice.

Lemma 12 (Balanced interval cuts). *Every nonempty equal-length pair of intervals in these two ordered vertex sets has at least two crossing edges, unless both intervals are the full sets.*

Proof. Every vertex degree is odd, so the cut is even as in Lemma 11. It suffices to exclude a zero cut for a proper pair. A zero cut selects entire stars. If a selected components are A -stars and b are B -stars, the numbers of selected rows and columns are $a + 3b$ and $3a + b$. Equal lengths imply $a = b \geq 1$.

The row interval crosses the junction between the two row blocks. Thus its A -centers form a suffix, and its B -leaves form a prefix. The reversed leaf order makes the selected B -center labels a suffix. In the column order the selected B -centers must instead form a prefix. A nonempty subset that is both a prefix and a suffix is the full set. Hence $b = q$, then $a = q$, and both intervals are full. $\square$

Choose any core edge (i, j) , delete its grid sensor $(i + 1, j + 1)$, and insert the two boundary sensors $(i + 1, 0)$ and $(0, j + 1)$. If $a, b \in \{0, 1\}$ indicate membership of row i and column j in an interval pair, the removed edge contributed $a \text{ xor } b$ to its cut and the two boundary sensors contribute $a + b$. Every cut weakly increases, and the formerly zero full cut gains exactly two. With the four corners the placement has $6q - 1 + 2 + 4 = 6q + 5$ sensors.

4.4 Without the corner requirement

Corners are not needed for robustness, but something on the boundary is. For a corner c of the grid, let Λ_c be the union of the two sides through c , with c itself removed.

Lemma 13 (Corner dipoles). *For $n \geq 2$, the dipole of the two neighbors of a corner c has detector set exactly Λ_c . Consequently every placement that survives one erasure for single targets has $|S \cap \Lambda_c| \geq 2$ for each corner, and contains at least four boundary sensors.*

Proof. Take $c = (0, 0)$ and the vertices $(0, 1)$ and $(1, 0)$. A sensor (x, y) has distances $x + |y - 1|$ and $|x - 1| + y$ to them. These agree when $x, y \geq 1$ and when $x = y = 0$; they differ by two when exactly one coordinate is zero. Each boundary sensor, corner or not, lies in exactly two of the four sets Λ_c , so $2(B + C) \geq 8$. $\square$

n	3	4	5	6	7	8	9	10	11
lower bound (Theorem 14)	3	4	6	8	9	10	12	14	15
$R_2(n)$	4	5	6	8	9	10	12	14	15
$R_2^\square(n)$	6	8	9	11	12	14	15	17	18
n	12	13	14	15	16	17	18	19	20
lower bound (Theorem 14)	16	18	20	21	22	24	26	27	28
$R_2(n)$	16	18	20	21	22	24	26	27	28
$R_2^\square(n)$	20	21	23	24	26	27	29	30	32

Table 1: One-erasure sensor budgets for two anonymous targets on the $n \times n$ grid. The corner-constrained optimum is $\lceil 3(n+1)/2 \rceil$ (Theorem 1); the unrestricted optimum equals the lower bound of Theorem 14 throughout the verified range $n \geq 5$.

Theorem 14 (Unrestricted lower bound). *For every $n \geq 3$,*

$$R_2(n) \geq \left\lceil \frac{3n-4}{2} \right\rceil + \varepsilon_n,$$

with ε_n as in Theorem 1.

Proof. Let S be robust. Theorem 8 gives (5), and Lemma 13 gives $B + C \geq 4$. Proposition 10 yields

$$4N \geq 6m + B + 4C \geq 6m + 4 + 3C \geq 6m + 4,$$

so $N \geq \lceil (3m+2)/2 \rceil = \lceil (3n-4)/2 \rceil$.

Now let $m = 4q$ with $q \geq 1$ and suppose $N = 6q + 1$. Then $4N = 6m + 4$, so $C = 0$, $B = 4$, and equality holds in (6). Since $N < 2m$, no interior line is empty, and the placement is ideal. In the notation of (8), adding the two line counts gives $4(\lambda + \nu) = 2m - 4$, so $\lambda + \nu = 2q - 1$ is odd; subtracting gives $\rho - \kappa = 2(\lambda - \nu) \equiv 2 \pmod{4}$. With $\rho + \kappa = 4$ this leaves $(\rho, \kappa) \in \{(3, 1), (1, 3)\}$. Suppose $\kappa = 1$. The half rows carry their sensors on the two sides $y = 0$ and $y = n - 1$, and the single half column carries its sensor on one of the sides $x = 0$, $x = n - 1$; the other of these two sides contains no sensor at all, since $C = 0$. The two corners on that empty side have sets Λ_c consisting of the empty side and one of the sides $y = 0$, $y = n - 1$ each. Lemma 13 therefore requires two sensors on each of those sides, so $\rho \geq 4$, a contradiction. The case $\rho = 1$ is symmetric. Hence $N \geq 6q + 2$. $\square$

Proposition 15 (Attainment in the verified range). *The bound of Theorem 14 is attained for every $5 \leq n \leq 20$, and $R_2(3) = 4$, $R_2(4) = 5$. Table 1 lists the values.*

Proof. For each $5 \leq n \leq 20$ the accompanying data contain an explicit corner-free placement of the stated size. Its robustness is verified from literal distance bags of all populations of mass at most two and all single erasures. Optimality is Theorem 14. For $n = 3, 4$ all smaller placements are enumerated. The placements were located by an integer-programming search, but neither their validity nor their optimality depends on a solver. $\square$

Conjecture 16. *For every $n \geq 5$, $R_2(n) = \lceil (3n-4)/2 \rceil + \varepsilon_n$.*

For $n \geq 5$ every optimal placement in the accompanying data is corner-free, with between four and six boundary sensors, and is built from three-leaf stars and half-lines as in Figure 1(b); for $n \not\equiv 2 \pmod{4}$ several are ideal in the sense of Proposition 10. What is missing for a proof is the analogue of Theorem 8 without corners. The dipoles of diagonal neighbors require two sensors in each pair of opposite quadrants at every split point, a condition the corners satisfy for

free; beyond these and the interval detectors there are further boundary families of two-target trades, whose detector sets contain neither an interval detector nor a dipole detector set: 12, 28, and 62 symmetry classes of inclusion-minimal sets for $n = 5, 6, 7$. They are large and were never binding in our searches: for $7 \leq n \leq 20$, the first placement of the lower-bound size meeting the interval and dipole constraints was already fully robust.

5 Rectangular geometry and additive-field topology

Write

$$G_{M,N} = P_{M+2} \square P_{N+2}, \quad V(G_{M,N}) = \{0, \dots, M+1\} \times \{0, \dots, N+1\}.$$

Thus M, N count interior coordinate lines, rather than vertices on a side. Let $R_2^{\text{corner}}(M, N)$ be the minimum number of sensors containing all four corners that recover every population of mass at most two after one arbitrary sensor erasure. Multiplicities and the empty population remain admissible. The coordinate distances below are the actual graph distances: a shortest path changes each coordinate monotonically, so its length is $|x - x'| + |y - y'|$.

Theorem 17 (Rectangular interval characterization). *A set S containing the four corners is robust in this sense if and only if*

$$|S \cap D(I, J)| \geq 2 \quad \text{for every nonempty pair of intervals } I \subseteq [M], \ J \subseteq [N] \text{ with } |I| = |J|,$$

where $D(I, J) = \{(x, y) : (x \in I) \text{ xor } (y \in J)\}$. *Necessity holds without requiring the corners.*

Proof. The skinny populations of Lemma 6 lie in this rectangle whenever $a + k \leq M$ and $b + k \leq N$. Their exact detector calculation is independent of the side lengths, proving necessity.

For sufficiency, opposite corner distances now sum to $M + N + 2$. Thus the two reports in each opposite pair contain equivalent information at fixed mass. At least one corner of each pair survives an erasure, and any difference detected at a corner is detected twice before erasure. Different masses differ at every sensor, and adjacent corners distinguish single vertices. It remains to consider distinct mass-two populations with equal corner reports. Their diagonal-coordinate bags agree just as in Lemma 7. Its canonical form has both coordinate spans $s + t$, where $s, t \geq 1$; the two open bounding intervals therefore have equal positive length $s + t - 1$. The actual rectangular bounds place the first in $[M]$ and the second in $[N]$. The same distance-sum calculation gives $D(I, J) \subseteq D(X, Y)$, so the interval hypothesis provides two detectors. If one interior coordinate set is empty, no such remaining collision can occur and the condition is correctly vacuous. $\square$

The singleton constraints retain their interpretation as rook-graph double total domination with boundary incidences. The numerical aspect-ratio pattern is already present in that literature: Kazemi et al. [21, Propositions 3.8–3.9] obtain $2n$ on an $n \times m$ rook board when $n \geq 2$ and $m > \lfloor (5n - 4)/3 \rfloor$, attained by two full columns. The $3/2$ square coefficient and three-leaf stars are also prior ingredients [21, 7]. Thus the ratio $5/3$ and the two-column idea are not new claims here. The following theorem transfers this pattern to the boundary-sensitive, corner-constrained sensing model and proves the additional all-interval recovery requirement.

Theorem 18 (Exact long-rectangle optimum and rigidity). *If $0 \leq N \leq M$ and $5N \leq 3M$, then*

$$R_2^{\text{corner}}(M, N) = 2N + 4.$$

The two boundary columns

$$S_{\partial} = \{0, M+1\} \times \{0, \dots, N+1\}$$

attain this value. If $5N < 3M$, every optimal placement has all the following properties:

1. Some line $x = i$, $1 \leq i \leq M$, contains no sensor.
2. Every line $y = j$, $1 \leq j \leq N$, contains exactly two sensors.
3. The only sensors on $y = 0$ or $y = N + 1$ are corners.
4. At most N of the M interior lines $x = i$ contain any sensor.

Proof. For the upper bound, the first coordinate of every sensor in S_∂ is outside every interior interval I . Hence

$$|S_\partial \cap D(I, J)| = 2|J| \geq 2.$$

Theorem 17 applies, and $|S_\partial| = 2(N + 2)$.

For the lower bound, form the bipartite incidence graph on the M interior x -lines and N interior y -lines. Interior sensors give edges, noncorner boundary sensors give half-edges, and the four corners have no incidence. Write $T = |S|$, let B count the half-edges, and denote line degrees by r_i, c_j . The singleton cuts require

$$r_i + c_j - 2x_{ij} \geq 2, \quad (*)$$

where x_{ij} indicates a sensor at (i, j) .

If all line degrees are positive, let L, Q, H count vertices of degrees 1, 2, ≥ 3 , and let h be the degree sum in the last class. A degree-one vertex incident with an ordinary edge has a neighbor of degree at least three by (*). Its edge can be charged to a distinct high-degree incidence. Other degree-one vertices use distinct half-edges. Therefore $L \leq h + B$ and $3H \leq h$. The total degree D satisfies $D = L + 2Q + h$, $D + B = 2(T - 4)$, and $L + Q + H = M + N$. Consequently

$$4(T - 4) - 3(M + N) = Q + (h + B - L) + (h - 3H) + B \geq B.$$

Since $3M \geq 5N$, this implies $T \geq 2N + 4$. If an x -line is empty, (*) forces every y -line to have degree at least two. These N lines contain disjoint sets of sensors and exclude the corners, giving $T \geq 2N + 4$. If a y -line is empty, the symmetric argument gives $T \geq 2M + 4 \geq 2N + 4$.

Now suppose $T = 2N + 4$ and $5N < 3M$. If every x -line were occupied, then either all y -lines would be occupied, contradicting the strict positive-line inequality, or an empty y -line would force $T \geq 2M + 4 > 2N + 4$. Thus an empty x -line exists. Every y -degree is at least two, and the $2N$ available noncorner sensors force equality on every y -line and leave no noncorner sensor on its boundary. Finally an occupied x -line cannot have degree one: its sensor lies on a y -line of degree two, making (*) read $1 + 2 - 2 \geq 2$. Each occupied x -line therefore uses at least two of the $2N$ noncorner sensors. There are at most N such lines. $\square$

This is an exact aspect-ratio regime, not a claim that $5/3$ is the precise transition for every pair of integer side lengths. The strict-case conclusion shows that optimal sensing can leave arbitrarily many complete coordinate lines uninstrumented without losing position recovery. The corner reports and interval constraints jointly supply the missing geometric information.

Scalar observations as a graph of potentials

Let R, C be finite nonempty coordinate sets of sizes A, B , and let K be a nontrivial abelian group. An *additive field* is

$$F(r, c) = a(r) + b(c) \in K.$$

For observed positions $E \subseteq R \times C$, form the bipartite graph H_E on *all* row and column vertices, with edge rc precisely when $(r, c) \in E$. Isolated vertices must be retained. Replacing the column potential by $-b(c)$ makes each field value a potential difference across an oriented edge. This is the classical graph-potential, or tension, viewpoint [23]. We include its proof and specialize it to distance moments; no priority claim is made for the general potential criterion or its elementary code-distance consequence.

Theorem 19 (Topology of additive-field reconstruction). *For arbitrary additive fields over K , the observations on E determine the entire field on $R \times C$ if and only if H_E is connected. The minimum number of sampling positions is $A + B - 1$, attained by any spanning tree. An arbitrary erased-edge pattern preserves recovery exactly when the remaining observation graph stays connected.*

An assignment $w : E \rightarrow K$ extends to an additive field if and only if its alternating sum vanishes on every cycle of H_E . It is enough to test the fundamental cycles of a spanning forest.

On the full $A \times B$ rectangle the minimum Hamming distance between distinct additive fields is exactly $\min(A, B)$. Thus precisely $e < \min(A, B)$ arbitrary erased entries can always be repaired.

Proof. Suppose two additive fields agree on E , and subtract their row and column potentials. Set $p(r) = a(r)$ and $p(c) = -b(c)$ on the bipartite vertex set. Agreement says $p(r) = p(c)$ along every observed edge. If H_E is connected, p is constant everywhere, and $a(r) + b(c) = 0$ on every position, proving uniqueness. Conversely, disconnectedness provides a row vertex r and a column vertex c in different components: if every row were connected to every column, nonemptiness of both parts would make the whole graph connected. Give the component of r a fixed nonzero potential $v \in K$ and all other components potential zero. The resulting field $p(r') - p(c')$ vanishes on E but is nonzero at (r, c) . This proves both necessity and the erasure-pattern statement. A connected graph on $A + B$ vertices has at least $A + B - 1$ edges, and every spanning tree attains that bound.

For consistency, orient each edge from its row to its column and seek $w(rc) = p(r) - p(c)$. Such differences telescope to zero around every cycle, with the signs alternating because the graph is bipartite. Conversely, choose one root and an arbitrary root potential in each component. Propagate potentials along a spanning tree using the prescribed edge values. Each remaining edge and its tree path form a fundamental cycle. Its alternating-sum condition says exactly that this edge also has the required potential difference. Hence all prescribed values extend.

For the Hamming bound, subtract two distinct fields and write their nonzero difference as $a(r) + b(c)$. If each row contains a nonzero entry, its support has at least A positions. Otherwise a zero row forces b to be constant. Every other row is then constant as a function of c , and a nonzero row gives at least B support positions. Thus the support has size at least $\min(A, B)$. Equality is attained by a field nonzero on one whole row or on one whole column, choosing whichever has fewer positions. The erasure claim follows because distinct words can be made identical by erasing all their differing positions, and by no smaller set containing that support. $\square$

The topology distinguishes two geometric events that a raw sensor count cannot: a spanning tree may determine the entire field from few values, while deleting all observations incident with a single coordinate vertex can destroy recovery even if nearly every other position is observed. In the potential representation, each connected component has an independent additive constant; a connected graph leaves just the harmless global change $a \mapsto a + t$, $b \mapsto b - t$.

Corollary 20 (Two full-bag anchors survive one erasure). *On any rectangle with at least two vertices in each direction, let every vertex report its first distance moment, and retain full distance bags at two opposite corners. Every population of mass at most two is recoverable after one arbitrary report erasure. If both sides have at least three vertices, the mass bound is sharp even with no erasures.*

Proof. For any population x , its moment field is additive:

$$M_x(i, j) = \sum_{u, v} x(u, v) |i - u| + \sum_{u, v} x(u, v) |j - v|.$$

A full bag also supplies its moment. The preceding theorem therefore repairs the one missing moment value, since both side sizes are at least two. At least one full corner bag survives and supplies the population mass h .

The reconstructed field determines coordinate marginals. If r_u is the marginal in the first coordinate, then

$$M_x(i+1, 0) - M_x(i, 0) = 2 \sum_{u \leq i} r_u - h.$$

All partial sums, and then the final marginal using total mass h , follow. The second coordinate is identical. A corner bag supplies the projection in a third direction, $u + v$ up to reflection and translation. These three directions recover every population of mass at most two by the projection argument in Proposition 25: a positive point of any reduced signed ambiguity would require negative points on three pairwise distinct lines through it, forcing negative mass at least three.

Sharpness is witnessed in a 3×3 subrectangle by the two mass-three populations

$$\{(0, 1), (1, 2), (2, 0)\}, \quad \{(0, 2), (1, 0), (2, 1)\}.$$

Their coordinate marginals and coordinate-sum bags coincide. Their moments agree everywhere, and their bags agree at both opposite corners of the ambient rectangle, so even the complete hybrid reports coincide. $\square$

The exact threshold in Theorem 19 applies to all additive fields. Feasible distance-moment fields form a more constrained class, so that theorem supplies a sufficient repair mechanism for those fields; its arbitrary-field converse is not claimed as a sharp population threshold. Corollary 20 trades a dense set of scalar reports for only two distribution-valued anchors, whereas Corollary 9 uses four anchors to permit sparse sensor sets.

The rectangular graph-distance bridge, complete interval equivalence, exact long-rectangle optimum, strict-case rigidity, additive-field connectivity equivalence, sharp arbitrary-field erasure threshold, and actual distance-moment decomposition and recovery implications are proved in the accompanying Lean modules. The general cycle-consistency and spanning-tree statements, and the final population-level assembly of Corollary 20, have the paper proofs above; these additional conclusions are not claimed as separate Lean endpoints.

6 Cartesian trades and compressed distance moments

A full distance bag may contain substantially more information than a scalar summary. This section derives exact capacity and fault thresholds by specializing classical spectral, null-design, interpolation, and projection methods to graph observations. The tensor decomposition and marginal conditions have their antecedents in Delsarte's Hamming-scheme theory and Martin's product-scheme designs [8, 26]. The cube extremizers below are also known Hamming functions. We retain these sensing corollaries for their precise contracts and reusable proofs; none is counted as an independently established novelty claim.

For a finite graph G , an integer population $X : V(G) \rightarrow \mathbb{N}$, and $k \geq 0$, put

$$M_k X(s) = \sum_{v \in V(G)} X(v) \text{dist}(s, v)^k, \quad 0^0 = 1.$$

Thus M_0 reports mass. In the results below, a k th-moment sensor reports only M_k , without separately reporting the lower moments or the mass.

Theorem 21 (Exact single-moment capacity). *Let $H = K_{q_1} \square \cdots \square K_{q_d}$, where $d \geq 1$ and $q_i \geq 2$. For every $k \geq 0$, two integer populations have the same k th-moment field on all vertices*

if and only if their marginals on every set of at most $\min(k, d)$ coordinates agree. If $k < d$, this observation recovers every population of mass at most $2^k - 1$, and fails for some two populations of mass 2^k . If $k \geq d$, it recovers populations of arbitrary mass.

Proof. Work first over $\mathbb{Q}$. In each coordinate, split the function space into constants and the subspace of functions with zero sum. Their tensor products give a direct-sum decomposition indexed by $J \subseteq [d]$; the coordinates in J carry the zero-sum factors. Let $j = |J|$. The coordinate shell-polynomial matrix is $t\mathbf{J}_{q_i} + (1-t)I$, so its eigenvalues on these two subspaces are $1 + (q_i - 1)t$ and $1 - t$, respectively. The product transform has eigenpolynomial

$$L_J(t) = (1-t)^j \prod_{i \notin J} (1 + (q_i - 1)t).$$

Applying $(t \frac{d}{dt})^k$ at $t = 1$ gives its k th-moment eigenvalue. Write the product after $(1-t)^j$ as $\sum_r c_r t^r$, with $c_r \geq 0$ and $c_0 = 1$. For the forward difference $\Delta f(r) = f(r+1) - f(r)$, the eigenvalue is

$$\lambda_J(k) = (-1)^j \sum_r c_r \Delta^j(r^k).$$

It vanishes for $j > k$. For $1 \leq j \leq k$, these differences are nonnegative and $\Delta^j(0^k) > 0$: inclusion–exclusion counts functions from a k -set to $r+j$ values that use each of the last j values. Hence the eigenvalue is nonzero with sign $(-1)^j$. The constant eigenvalue is also positive: for $k = 0$ it is $|V(H)|$, and for $k > 0$ the distance distribution has a positive-distance term because $d \geq 1$ and $q_i \geq 2$. The kernel therefore consists exactly of the interactions of order greater than k . Marginalization to I kills precisely the components with $J \not\subseteq I$, proving the stated observation equivalence.

We recall the integer trade bound with its short proof, allowing repeated points. A nonzero integer array whose marginals on at most k coordinates vanish has positive mass at least 2^k . For $k = 0$, this is immediate from zero total sum. Induct on the dimension. For $k \geq 1$, every nonzero slice obtained by fixing one coordinate is a $(k-1)$ -trade. If at least two slices are nonzero, their positive masses sum to at least $2 \cdot 2^{k-1}$. If only one is nonzero, it is itself a k -trade in one fewer coordinate. When k reaches the dimension, the full marginal is the array itself, so no nonzero trade exists. This proves the bound and the recovery assertion.

For sharpness when $k < d$, choose two values in each of $k+1$ coordinates and fix all remaining coordinates. The even- and odd-parity corners each have mass 2^k and identical marginals on at most k coordinates. Adding a common population gives ambiguity at every greater prescribed mass. Finally, when $k \geq d$, the full-coordinate marginal is known. $\square$

The minimum volume 2^k and the parity construction are classical. Frankl and Pach [11, Theorem 1 and Proposition 2] give the null-design support and alternating-trace arguments; with integer coefficients these imply the positive-mass bound used here. See also Ghorbani et al. [12, Lemma 4]. The spectral calculation specializes classical Hamming and product-scheme methods to a *single* raw-distance moment: its extra step is checking that no eigenvalue of order at most k vanishes. Together with the trade bound this yields the exact population capacity. Unequal alphabets do not constitute a new marginal theory; product-scheme incidence and spectral conditions already appear in Martin [26, Theorem 2.3].

Theorem 22 (Optimal sampling of a cube moment field). *Let $d \geq 1$ and $0 \leq k \leq d$. The k th-moment values at the*

$$N_{d,k} = \sum_{j=0}^k \binom{d}{j}$$

vertices of Hamming weight at most k determine the entire k th-moment field on Q_d . No set of fewer scalar sensors can determine every such field of arbitrary integer populations.

Proof. The eigenvalue computation above identifies the range of M_k with the space of multilinear polynomials of degree at most k on $\{0, 1\}^d$. Write $f(s) = \sum_{|I| \leq k} a_I \prod_{i \in I} s_i$. The samples in the lower Hamming ball give

$$a_I = \sum_{J \subseteq I} (-1)^{|I|-|J|} f(\mathbf{1}_J), \quad |I| \leq k,$$

by Boolean-lattice inversion, and therefore determine f . The range has dimension $N_{d,k}$. Fewer evaluations have a nonzero rational kernel in this range. A nonzero invisible field has a rational preimage under M_k ; clearing denominators and splitting that preimage into its positive and negative parts produces two integer populations with equal sampled reports but different complete fields. $\square$

The lower bound here concerns arbitrary population masses and reconstruction of the whole field. It is not a lower bound on the separate problem of minimizing sensors for a fixed small mass. Hamming balls as interpolation sets, and the dimension $N_{d,k}$, are standard Reed–Muller ingredients; see the preliminaries of Gopalan et al. [15].

Theorem 23 (Sharp robustness of compressed cube reports). *Suppose $d \geq 2k \geq 2$ and $2^{k-1} \leq h \leq 2^k - 1$. The minimum Hamming distance between all-sensor k th-moment reports of distinct populations of mass at most h on Q_d is exactly 2^{d-k} . Consequently universal recovery from e arbitrary report errors and u known erasures holds exactly when*

$$2e + u < 2^{d-k}.$$

The same minimum distance holds for populations of exact mass h . For $h \geq 2^k$, the minimum distance is zero.

Proof. Below mass 2^k , Theorem 21 makes the difference field nonzero. It is a multilinear polynomial of degree at most k , so it is nonzero at at least 2^{d-k} Boolean vertices. Here is the usual characteristic-zero version of the polynomial weight argument. Write $f = a + x_d b$. If $b = 0$, the equal slices double the induction bound. If $b \neq 0$, it has degree at most $k - 1$, and at every point where $b \neq 0$ at least one of $a, a + b$ is nonzero. Induction gives the same bound; the endpoint $k = d$ is immediate from nonempty support.

For attainment, use the paired-coordinate construction of Valyuzhenich [33, Section 4, Lemma 5]. Partition the first $2k$ coordinates into k pairs. On each pair use the signed population $a_i = \delta_{(0,1)} - \delta_{(1,0)}$, fix the remaining coordinates at zero, and put $z = a_1 \otimes \cdots \otimes a_k$. Its two signs each have mass 2^{k-1} . Multinomial expansion gives

$$M_k z(s) = k! \prod_{i=1}^k (\text{dist}(s_i, (0, 1)) - \text{dist}(s_i, (1, 0))) = k! \prod_{i=1}^k 2(s_{2i-1} - s_{2i}).$$

Terms omitting one factor vanish because that factor has zero mass. This field is nonzero exactly when the two bits in every selected sensor pair differ. There are $2^k 2^{d-2k} = 2^{d-k}$ such sensors. Common padding reaches exact mass h . The usual errors-and-erasures distance criterion proves the claimed recovery threshold: if the distance is at most $2e + u$, erase some differing positions and divide the remaining ones between two error patterns of size at most e . The parity trade from Theorem 21 gives distance zero at and above mass 2^k . $\square$

Both tensor shapes are already explicit in Valyuzhenich [33, Section 4]: in that notation the input is a member of $F_1(d, k, d - k)$, and its moment output is proportional to a member of $F_1(d, k, k)$. Their supports are 2^k and 2^{d-k} . The source works with real-valued Hamming functions, so no finite-field analogy is required. Its Theorem 3 also gives the report-support lower bound directly on the spectral interval $U_{[0,k]}$. The calculation above is a sharp sensing corollary: it translates the known input into two nonnegative integer populations and the output support into a report distance. It does not introduce a new extremal tensor geometry.

Theorem 24 (Tensor detector and moment propagation). *For a finite connected graph G and signed integer population a , define*

$$R_s(a; T) = \sum_v a(v) T^{\text{dist}(s,v)}, \quad D_G(a) = \{s : R_s(a; T) \neq 0\}.$$

For finite connected graphs G, H and signed populations a, b ,

$$R_{(s,t)}(a \otimes b; T) = R_s(a; T) R_t(b; T), \quad D_{G \square H}(a \otimes b) = D_G(a) \times D_H(b).$$

Thus the Hamming distance of this shell-report trade on a Cartesian sensor set is the product of its factor distances. If a, b are balanced with positive masses A, B , the tensor has positive mass $2AB$. If their first nonzero moments at the selected sensors have orders p, q , the tensor's first nonzero order is $p + q$, with value

$$\binom{p+q}{p} M_p a(s) M_q b(t).$$

Proof. Cartesian-product distances add, so expanding the response polynomial proves the factorization. Since $\mathbb{Z}[T]$ has no zero divisors, its nonvanishing set is exactly the Cartesian product. The tensor has total sum zero and ℓ_1 norm $(2A)(2B)$, proving its positive-mass formula. For each j , binomial expansion gives

$$M_j(a \otimes b)(s, t) = \sum_{i=0}^j \binom{j}{i} M_i a(s) M_{j-i} b(t).$$

For $j < p + q$ every summand has a vanishing factor. At $j = p + q$, only $i = p$ can survive, and its displayed value is nonzero. $\square$

Iterating the theorem on any product of d nontrivial connected graphs gives a universal compression obstruction: choose two vertices in each factor and take the tensor of their differences. Its two signs have mass 2^{d-1} , and every moment of degree below d vanishes at every sensor. The d th moment is $d!$ times the product of the factor distance contrasts. This is the distance-sensing form of the classical mixed finite-difference construction. Polynomial switching components and projected hypercube trades have extensive antecedents in discrete tomography [16]; the exact detector and fault propagation formulas above specify their consequences for Cartesian graph shell reports. The factorization, integral-domain nonvanishing, and binomial moment identity are standard algebraic operations, packaged here as reusable sensing tools.

Proposition 25 (Sharp hybrid capacities on rectangles). *Suppose a rectangular grid with at least two vertices in each direction reports its first-distance moment at every vertex. Adding one full corner bag recovers every population of mass at most two; this fails at mass three if both side lengths are at least three. Adding two adjacent full corner bags recovers every population of mass at most three; this fails at mass four if both side lengths are at least four. Opposite corner bags are equivalent, so using all four corners does not improve this capacity when both sides are at least four. It suffices to sample the scalar field on one row and one column through a corner.*

Proof. Use coordinates $0 \leq x < a$, $0 \leq y < b$. A full bag gives the mass h . Writing r_x for the row marginal, successive bottom-boundary values satisfy

$$M_1(i+1, 0) - M_1(i, 0) = 2 \sum_{x \leq i} r_x - h.$$

These differences, together with h , determine every row marginal; the left-boundary values similarly determine the column marginal. Conversely those two marginals determine the scalar

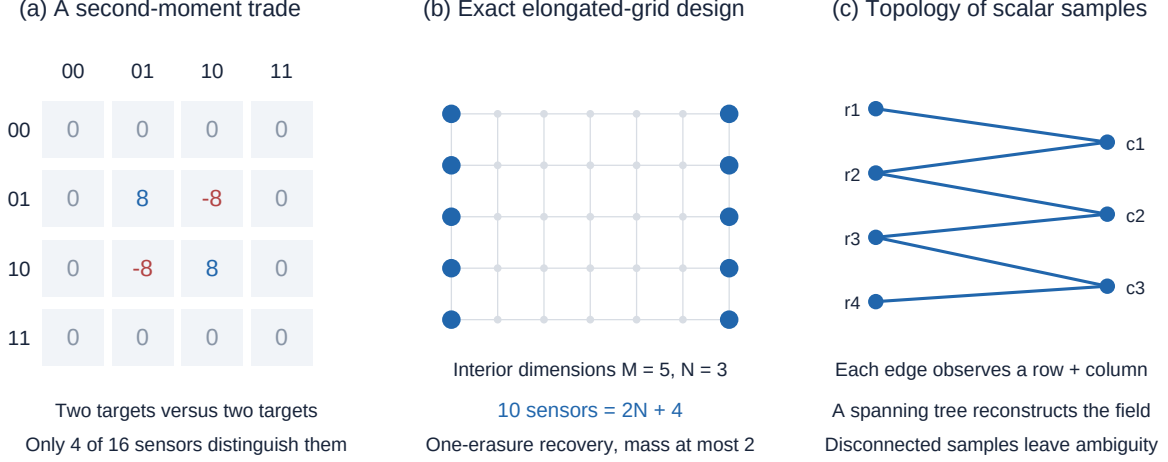


Figure 3: Three sensing mechanisms. (a) The second-moment difference on Q_4 for the trade $\{0101, 1010\} - \{0110, 1001\}$; rows and columns index the two coordinate pairs. Its four nonzero reports attain the sharp distance bound. (b) The two boundary columns, including all corners, give the exact ten-sensor design on a 7×5 rectangle. (c) Sampling an additive field at the edges of a spanning tree determines every row–column sum.

field everywhere. The corner $(0, 0)$ reports the $x + y$ projection. The adjacent corner $(0, b - 1)$ reports the $x - y$ projection up to a fixed shift. The two contracts are therefore exactly the three projections $x, y, x + y$ and the four projections $x, y, x + y, x - y$.

For r pairwise nonparallel projection-line families, a nonzero reduced integer trade has negative mass at least r : choose a positive point. Each of the r lines through it must contain a negative point to balance its projection. These lines intersect only at the chosen positive point, so the required negative points are distinct. This is the classical projection-uniqueness argument underlying the finite-support theorem proved by Hajós and published by Rényi [27]; modern weighted versions include Fraiman et al. [10, Theorems 1.1 and 2.1]. It proves the two recovery assertions without a simplicity assumption. The capacities are applications of this projection theory once the scalar field has supplied the coordinate marginals.

Sharp witnesses for the respective contracts are

$$\{(0, 1), (1, 2), (2, 0)\} \quad \text{and} \quad \{(0, 2), (1, 0), (2, 1)\},$$

and

$$\{(3, 2), (2, 0), (1, 3), (0, 1)\} \quad \text{and} \quad \{(3, 1), (2, 3), (1, 0), (0, 2)\}.$$

Their three or four corresponding projection multisets agree by direct inspection. The four-direction switching component is also the coefficient array of $(X - 1)(Y - 1)(X - Y)(XY - 1)$, within the classical polynomial framework of Hajdu and Tjeldeman [16]. An opposite corner replaces each reported distance r by $a + b - 2 - r$, so supplies no additional projection. Larger rectangles contain the same witnesses, and common padding gives failures at all larger masses. $\square$

The new formal modules verify tensor shell factorization, the detector equivalence, binomial moment propagation, and abstract Boolean-polynomial interpolation on a lower Hamming ball. The graph moment-range bridge, integer trade bound, complete capacity and robustness theorems, and the hybrid projection argument have complete paper proofs but are not unconditional Lean endpoints. Section 12 distinguishes these bridges from the checked algebra and the selected comparison statements.

7 Blindness: the equidistant dimension of hypercubes

The third question of the calculus asks for sensors that are *blind*. By (2), a sensor s is blind to the dipole $\delta_x - \delta_y$ exactly when $d(s, x) = d(s, y)$. A *distance-equalizer set* S of a graph has, for every two distinct vertices $x, y \notin S$, a member blind to their dipole; its minimum cardinality is the equidistant dimension $\xi(G)$ [14]. Resolving sets meet every dipole detector set, while equalizers meet every dipole blind set, so the two parameters are the mass-one extremes of the same detector calculus.

In the n -cube Q_n , vertices are binary words and graph distance is Hamming distance. Let E_n, O_n be the even- and odd-weight words, write $\bar{x}$ for the bitwise complement of x , and let $\sigma(x) \in \{-1, 1\}^n$ be the sign vector $\sigma(x)_i = (-1)^{x_i}$. The first-moment report of the antipodal dipole is a linear form on the sign cube:

$$M_1(\delta_{\bar{b}} - \delta_b)(y) = d(y, \bar{b}) - d(y, b) = n - 2d(y, b) = \langle \sigma(b), \sigma(y) \rangle. \quad (13)$$

A sensor is therefore blind to an antipodal dipole exactly on a hyperplane through the origin, and products of these first-moment fields are the polynomials used below.

If $4 \nmid n$, an entire parity class is a minimum equalizer and $\xi(Q_n) = 2^{n-1}$ [13, 24]. The remaining dimensions are the subject of this section. Gispert-Fernández, Rodríguez-Velázquez, and Yero [13, Conjecture 1] conjectured

$$\xi(Q_n) = 2^{n-1} + 2^{n/2-2} \quad (4 \mid n). \quad (14)$$

We replace the exponential correction by a linear one and determine it up to a single unit.

Theorem 26 (Hypercube equalization within one). *For every positive integer n divisible by four,*

$$2^{n-1} + \frac{n}{2} - 1 \leq \xi(Q_n) \leq 2^{n-1} + \frac{n}{2}.$$

The lower value is attained for $n = 4$ and the upper value for $n = 8, 12, 16$:

$$\xi(Q_4) = 9, \quad \xi(Q_8) = 132, \quad \xi(Q_{12}) = 2054, \quad \xi(Q_{16}) = 32776.$$

In particular (14) fails for every $n \geq 12$ divisible by four, by an exponentially large margin.

The theorem assembles Theorems 27, 30, 35, and 39 below. Together with the cited result for $4 \nmid n$, the equidistant dimension of every hypercube is now known up to an additive error of at most one, and exactly in every dimension below twenty.

7.1 The linear upper bound and the folded graph

Theorem 27 (Linear antipodal correction). *For every positive integer q , putting $n = 4q$ gives*

$$\xi(Q_n) \leq 2^{n-1} + \frac{n}{2}.$$

In particular, $\xi(Q_{12}) \leq 2054 < 2064$, contradicting (14). The inequality is strict against the conjectured value for every $4 \mid n$ with $n \geq 12$.

The construction comes from prefix balancing, a classical idea in balanced coding [22, 2]. Its application to this equidistance conjecture is the point here; we do not claim the prefix-flip technique itself as new. The necessity of selecting an entire bipartition class is already known [14, 13], and the shared-coordinate parity argument below appears in the proof of Theorem 3.1 of [13].

Lemma 28 (Parity of a cube bisector). *Let x, y have even Hamming distance $h > 0$. Their bisector has*

$$2^{n-h} \binom{h}{h/2}$$

vertices. If $h < n$, exactly half belong to each parity class. If $h = n$, all bisectors have parity $|x| + n/2 \pmod{2}$.

Proof. On the h differing coordinates a bisector must agree with x on exactly $h/2$ positions. The other $n - h$ coordinates are free. If one is free, toggling it pairs the two parity classes. When none is free, every bisector is at distance $n/2$ from x , giving the stated parity. $\square$

Thus, for $4 \mid n$, the odd class contains a blind sensor for every nonantipodal dipole of even vertices of even vertices. Only antipodal dipoles remain. This observation gives an exact transformation, not just an upper-bound construction.

Definition 29. *The folded even orthogonality graph Ω_n , for positive n with $4 \mid n$, has vertices the antipodal classes $[x] = \{x, \bar{x}\}$ with $x \in E_n$. Two different classes are adjacent when $d(x, y) = n/2$. Complementing either representative preserves this condition.*

This is the even component of the antipodal twin quotient studied by Boutin and Cockburn [5, Section 4]. The graph and its order and degree are known; the following statement concerns its ordinary domination number and the equidistant problem.

Theorem 30 (Exact antipodal reduction). *For positive n with $4 \mid n$,*

$$\xi(Q_n) = 2^{n-1} + \gamma(\Omega_n),$$

where γ is ordinary closed-neighborhood domination.

Proof. In any connected bipartite graph, two vertices in opposite parts have no equidistant vertex: their distances from any fixed vertex have different parity. A distance-equalizer set must therefore contain an entire part. By toggling a fixed coordinate, assume this is O_n and write $S = O_n \cup T$, with $T \subseteq E_n$.

Lemma 28 handles all nonantipodal pairs outside S . For an antipodal pair $[x]$, the condition is precisely that either T contains an endpoint, or some $t \in T$ is at distance $n/2$ from x . This says that the projection of T is a dominating set in Ω_n . Its size is at most $|T|$, proving the lower bound. Conversely, lift one representative of each class in a dominating set of Ω_n ; the same argument verifies that its union with O_n is a distance-equalizer set of the required size. $\square$

The quotient has 2^{n-2} vertices, is vertex-transitive under even translations, and is regular of degree $\frac{1}{2} \binom{n}{n/2}$: a fixed representative has $\binom{n}{n/2}$ middle-distance words, all even, two in each neighboring class. It separates the unavoidable parity cost from the remaining domination problem.

For $0 \leq j \leq n/2$, let p_j be the word whose first $2j$ bits are one and whose other bits are zero. Define

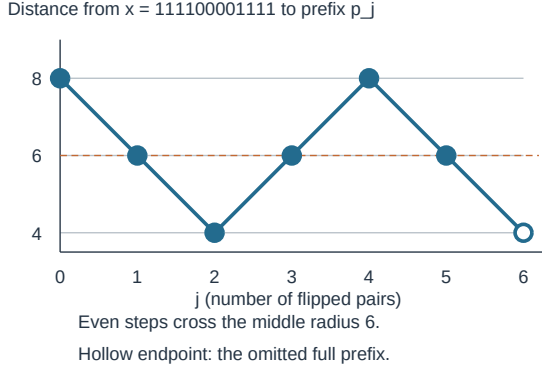
$$T_n = \{p_0, p_1, \dots, p_{n/2-1}\}.$$

All these $n/2$ distinct words are even.

Lemma 31 (Even-prefix balance). *Every $x \in E_n$, with positive n and $4 \mid n$, has $d(x, p_j) = n/2$ for some $0 \leq j < n/2$.*

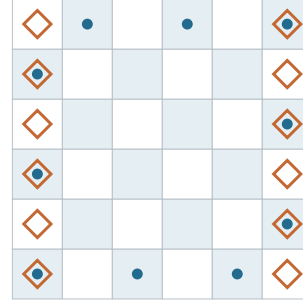
Proof. The sequence $f(j) = d(x, p_j)$ starts at $|x|$ and ends at $n - |x|$. Each step flips two bits, so $f(j+1) - f(j) \in \{-2, 0, 2\}$. Every value is even, and $n/2$ is even. The endpoints lie on opposite sides of $n/2$, so a discrete intermediate-value argument supplies an equality. If the last endpoint is the first equality obtained, then $|x| = n/2$, and the initial endpoint already works. The final prefix can therefore be omitted. $\square$

(a) Prefix balance in dimension 12



The cube argument uses a crossing along a path;
the bishop argument uses a fixed-point-free pairing.

(b) Boundary parity on a 6 by 6 board



Blue dots: 10 even boundary squares.
Diamonds: 12 bishops give the upper bound.
Equality would pair diagonal counts:
 $p_0 = p_{10}, p_2 = p_8, p_4 = p_6$.

Figure 4: Two geometric mechanisms. Left: an even-valued distance function crosses the middle radius along a prefix path. Right: the bishop lower bound pairs diagonal populations on opposite boundaries, while the two outer columns realize the upper bound.

Taking $S = O_n \cup T_n$ proves Theorem 27 and the upper bound $\gamma(\Omega_n) \leq n/2$. In dimension twelve the six additional vertices, with the least significant bit interpreted as the first coordinate, are

$$0, \quad 3, \quad 15, \quad 63, \quad 255, \quad 1023.$$

The proof is uniform in the dimension. The finite verification additionally checks all $\binom{2042}{2} = 2,083,861$ pairs outside this explicit 2054-vertex set, using independently constructed bisectors.

Geometric flexibility. The prefix path may flip coordinates in any order and may start at any even vertex. Its even-indexed points still form a middle-sphere cover of the even class. The mechanism is a path from a vertex to its antipode, with a parity-restricted distance function crossing the middle radius; it does not depend on the numerical binary labels.

7.2 The matching lower bound

Say that an even word x is *balanced* by a set $B \subseteq E_n$ if $d(x, b) = n/2$ for some $b \in B$, and write $M(B)$ for the set of even words that B does not balance. Since $d(\bar{x}, b) = n - d(x, b)$, the set $M(B)$ is closed under complementation; let $\mu(B) = |M(B)|/2$ count its antipodal classes. By (13), the product of first-moment dipole fields

$$f_B(y) = \prod_{b \in B} \langle \sigma(b), y \rangle, \quad y \in \{-1, 1\}^n, \quad (15)$$

vanishes at $\sigma(x)$, $x \in E_n$, exactly when B balances x . Two elementary facts control it. Write $\chi(y) = y_1 y_2 \cdots y_n$, so that $\chi(\sigma(x)) = (-1)^{|x|}$.

Lemma 32 (Parity orthogonality). *If P is a real polynomial in $y_1, \dots, y_n$ of total degree less than n , then*

$$\sum_{y \in \{-1, 1\}^n} \chi(y) P(y) = 0;$$

equivalently, the sums of P over the even and over the odd sign vectors agree.

Proof. By linearity take P to be a monomial. On the sign cube it equals a squarefree monomial $\prod_{i \in I} y_i$ with $|I| < n$. Choose $j \notin I$. Negating y_j preserves the monomial and negates χ , so the terms cancel in pairs. $\square$

Lemma 33 (Odd points are never balanced). *Let $4 \mid n$. If b is even and x is odd, then $\langle \sigma(b), \sigma(x) \rangle \equiv 2 \pmod{4}$. Hence f_B has no zero at an odd point when $B \subseteq E_n$.*

Proof. The distance $d(b, x)$ is odd, and $\langle \sigma(b), \sigma(x) \rangle = n - 2d(b, x)$. $\square$

Lemma 32 is the degree form of the parity lemma of Alon, Bergmann, Coppersmith, and Odlyzko [2, Lemma 2.1], who used it to prove that $n/2$ even vectors are needed to balance all of E_n . The next theorem measures how badly fewer vectors must fail.

Theorem 34 (Quantitative balancing). *Let $4 \mid n$, let $0 \leq t \leq n/2 - 1$, and let $B \subseteq E_n$ have $|B| = n/2 - 1 - t$. Every real polynomial of degree at most t that vanishes on $\sigma(M(B))$ vanishes on the whole sign cube. Consequently*

$$\mu(B) \geq \max \left\{ \sum_{\substack{0 \leq j \leq t \\ j \text{ even}}} \binom{n}{j}, \sum_{\substack{0 \leq j \leq t \\ j \text{ odd}}} \binom{n}{j} \right\}. \quad (16)$$

In particular $\mu(B) \geq 1$ when $|B| = n/2 - 1$ and $\mu(B) \geq n$ when $|B| \leq n/2 - 2$. Equality holds in (16) when $|B| \leq 1$.

Proof. Let q have degree at most t and vanish on $\sigma(M(B))$. The polynomial $(f_B q)^2$ has degree at most $2(n/2 - 1 - t) + 2t = n - 2$. It vanishes at every even point: f_B vanishes at the balanced ones and q at the others. Lemma 32 therefore gives $\sum_{y \text{ odd}} (f_B q)^2(y) = 0$, so $f_B q$ vanishes at every odd point, and Lemma 33 shows that q does. Now q^2 has degree at most $2t \leq n - 2$ and vanishes at all odd points; a second application of Lemma 32 makes it vanish at all even points as well.

For $p \in \{0, 1\}$, the squarefree monomials of degree at most t and parity p are linearly independent functions on the sign cube. Each function q in their span satisfies $q(-y) = (-1)^p q(y)$, so its restriction to $\sigma(M(B))$ is determined by its values on one representative of each antipodal class. The first part makes this restriction injective, and comparing dimensions proves (16).

If $B = \emptyset$, then $t = n/2 - 1$ is odd, $\mu = 2^{n-2}$, and the odd binomial coefficients below the middle also sum to 2^{n-2} . If $B = \{b\}$, then $t = n/2 - 2$ is even and both sides equal $2^{n-2} - \frac{1}{2} \binom{n}{n/2}$. $\square$

For comparison, the hyperplane-covering theorem of Alon and Füredi [1, Corollary 1], applied to the doubled family $B \cup (B \oplus e_1)$, gives only $\mu(B) \geq 4^t$. The polynomial dimension count in (16) is what makes the following consequence possible.

Theorem 35 (Folded domination within one). *For every positive n divisible by four,*

$$\frac{n}{2} - 1 \leq \gamma(\Omega_n) \leq \frac{n}{2}.$$

Moreover, if D is a dominating set with fewer than $n/2$ vertices, then $|D| = n/2 - 1$ and the induced graph $\Omega_n[D]$ has an isolated vertex.

Proof. The upper bound is Lemma 31. Let D dominate Ω_n , and lift one representative of each class to $B \subseteq E_n$. An unselected class is adjacent to a selected one, so its lifts are balanced. A selected class has a balanced lift exactly when it has a selected neighbor. Hence $M(B)$ consists of the lifts of the isolated vertices of $\Omega_n[D]$, and

$$\mu(B) = \#\{\text{isolated vertices of } \Omega_n[D]\} \leq |D|. \quad (17)$$

If $|D| \leq n/2 - 2$, Theorem 34 gives $\mu(B) \geq n > |D|$, a contradiction. If $|D| = n/2 - 1$, it gives $\mu(B) \geq 1$. $\square$

The argument has a direct form, which is the one checked in Lean. Suppose $|D| \leq n/2 - 2$ and choose a nonzero $v \in \mathbb{R}^n$ orthogonal to the fewer than n vectors $\sigma(b)$, $b \in B$. The polynomial $f_B(y)^2 \langle v, y \rangle^2$ is a product of at most $n-2$ linear forms and vanishes at every even point, because an unbalanced even word is a lift of a selected class. Parity orthogonality makes it vanish at every odd point, Lemma 33 then gives $\langle v, y \rangle = 0$ for all odd y , and the odd sign vectors span $\mathbb{R}^n$.

Corollary 36 (Total and paired domination). *For every $q \geq 1$, the $2q$ prefix classes in Ω_{4q} form a total dominating set whose induced graph is q disjoint edges, and*

$$\gamma_t(\Omega_{4q}) = \gamma_{\text{pr}}(\Omega_{4q}) = 2q.$$

Proof. Distinct prefixes in the stated range represent distinct antipodal classes. Their distances satisfy $d(p_i, p_j) = 2|i - j|$, so the induced edges are exactly $[p_i][p_{i+q}]$ for $0 \leq i < q$. Every class, including each selected class, has an orthogonal selected prefix by Lemma 31; this is a distinct class because its distance is $2q$. Thus the set is totally dominating and its induced graph has a perfect matching. A total dominating set D has $\mu(B) = 0$, so the case $t = 0$ of Theorem 34 forces $|D| \geq n/2$; this is the balancing theorem of [2, Theorem 2, $d = 0$]. Paired domination implies total domination. $\square$

Total domination therefore costs exactly $n/2$, and ordinary domination can save at most one vertex, only by leaving a selected class isolated. At that boundary the dipole product is rigid.

Proposition 37 (Rigidity at deficit one). *Let D dominate Ω_n with $|D| = n/2 - 1$, lifted to B , and let $U = M(B)$ be the lifts of its isolated classes. Then f_B is determined by its values on U : for every y in the sign cube,*

$$f_B(y) = 2^{1-n} \sum_{u \in U} f_B(\sigma(u)) \sum_{|S| < n/2} \prod_{i \in S} \sigma(u)_i y_i.$$

Proof. Write $\hat{f}(S) = 2^{-n} \sum_y f_B(y) \prod_{i \in S} y_i$ for the coefficients of f_B in the squarefree monomial basis; they vanish for $|S| \geq n/2$ since $\deg f_B < n/2$. The function $g = \frac{1}{2}(1 + \chi)f_B$ agrees with f_B at even points and vanishes at odd points, so it is supported on $\sigma(U)$. Its coefficients are $\hat{g}(S) = \frac{1}{2}(\hat{f}(S) + \hat{f}(S^c))$, which equals $\frac{1}{2}\hat{f}(S)$ when $|S| < n/2$. Hence $\hat{f}(S) = 2\hat{g}(S) = 2^{1-n} \sum_{u \in U} f_B(\sigma(u)) \prod_{i \in S} \sigma(u)_i$ for $|S| < n/2$, and summing the expansion of f_B gives the formula. $\square$

The proposition says that a hypothetical dominating set of size $n/2 - 1$ makes a product of $n/2 - 1$ linear forms equal to a combination of at most $n/2 - 1$ truncated point masses at its own isolated members. For $n = 4$ this happens: $B = \{0\}$, $f_B(y) = y_1 + y_2 + y_3 + y_4$, and $\Omega_4 = K_4$. The computations below show that it does not happen for $n = 8, 12, 16$.

7.3 Three common middle neighbors

The isolated classes of a small dominating set can always be repaired three at a time.

Theorem 38 (Three common middle neighbors and a four-set obstruction). *For positive n divisible by four, every set of at most three vertices of Ω_n has a common neighbor. The four classes represented by*

$$0, \quad e_1 \oplus e_2, \quad e_1 \oplus e_3, \quad e_2 \oplus e_3$$

have no common neighbor. Thus they form a minimal family of open neighborhoods with empty intersection.

Proof. By an even translation, take the three representatives to be $0, a, b$. Partition the coordinates into four blocks indexed by $(a_i, b_i) \in \{0, 1\}^2$, with sizes $c_{00}, c_{01}, c_{10}, c_{11}$. Since a, b and n are even, all four sizes have the same parity. If they are even, select half of each block. If they are odd, select $(c_{00} + 1)/2$ and $(c_{11} + 1)/2$ from the diagonal blocks, and $(c_{01} - 1)/2$ and $(c_{10} - 1)/2$ from the off-diagonal blocks. The resulting word z satisfies

$$|z| = n/2, \quad |z \cap a| = |a|/2, \quad |z \cap b| = |b|/2.$$

Consequently its distances to $0, a, b$ are all $n/2$. It is even, and its class is a common neighbor. Repeating a representative handles nonempty smaller sets; the empty family has every vertex as a common neighbor.

For the displayed four-set, a common middle point would satisfy $|z| = n/2$ and

$$z_1 + z_2 = z_1 + z_3 = z_2 + z_3 = 1.$$

The equations force $z_1 = z_2 = z_3$, then $2z_1 = 1$, impossible for a bit. The four classes are distinct; every proper subfamily has a common neighbor by the first part. This is a minimal four-set obstruction, not a claim that four-wise intersection implies global intersection. $\square$

In dimension eight the obstruction is itself extremal: the four classes of Theorem 38 are pairwise nonadjacent and dominate Ω_8 , so Ω_8 has an *independent* minimum dominating set. In dimension twelve the opposite holds; see Theorem 39.

Earlier route. A previous version of this work combined Theorem 38 with the Alon–Füredi inequality to obtain $2^{n-2|D|} \leq 4r$ for a dominating set with r isolated selected classes, and hence $\gamma(\Omega_n) \geq n/2 - \frac{1}{2} \log_2(2n)$. Theorem 35 supersedes that estimate. The finite support-certificate modules written for it remain in the Lean library but are no longer needed for any statement in this paper. Near- $n/2$ balancing bounds with different quantifiers appear in Hrubeš et al. [20, Theorems 4–5]: every target of weight $n/2$ must bisect a selected proper nonempty subset. Here the targets have all even weights and selected classes may be exceptions, so those results do not imply the domination statement; see also [3] for polynomial support bounds on the cube.

7.4 Exact values through dimension sixteen

Theorem 35 leaves one question in each dimension: does a dominating set of size $n/2 - 1$ exist? The quotient and the isolate structure make that a finite calculation of manageable size.

Theorem 39 (Exact small values; exhaustive computation for $n \geq 8$).

$$\gamma(\Omega_4) = 1, \quad \gamma(\Omega_8) = 4, \quad \gamma(\Omega_{12}) = 6, \quad \gamma(\Omega_{16}) = 8.$$

Consequently $\xi(Q_4) = 9$, $\xi(Q_8) = 132$, $\xi(Q_{12}) = 2054$, and $\xi(Q_{16}) = 32776$. Moreover:

1. every set of three vertices of Ω_8 leaves at least 4 vertices undominated, and every set of five vertices of Ω_{12} leaves at least 20; both bounds are attained;
2. Ω_8 has independent minimum dominating sets, whereas every minimum dominating set of Ω_{12} is a total dominating set.

Reduction and computational proof. The graph Ω_4 is K_4 . The prefixes give the upper bounds, so it remains to exclude dominating sets of size $n/2 - 1$ for $n = 8, 12, 16$.

Common reductions. Translation by an even word preserves parity, complementation, and Hamming distances, so one selected representative may be taken to be the zero word. A permutation of the n coordinates preserves the covering property. Arrange t further representatives as the rows of a $t \times n$ binary matrix; it suffices to consider matrices whose columns are sorted by

their t -bit pattern, that is, weak compositions of n into 2^t pattern counts. The even words with a fixed coordinate equal to zero represent the 2^{n-2} classes, and a representative x is covered by a center b precisely when $|x \oplus b| \in \{0, n/2, n\}$.

Dimensions eight and twelve. For $n = 8$ all $\binom{63}{2}$ pairs completing the zero class are inspected. For $n = 12$ there are $\binom{27}{15} = 17,383,860$ compositions for $t = 4$, of which exactly

$$\frac{1}{16} \left(\binom{27}{15} + 15 \binom{13}{7} \right) = 1,088,100$$

have four even rows, by averaging the sixteen parity characters. An exact-integer program enumerates all of them and finds a minimum of twenty uncovered classes, attained by the centers 0, 3510, 3591, 3640, 4032 in integer binary encoding. A second implementation enumerates the tuples by stars and bars, chooses representatives with the first coordinate zero, and tests literal distances.

Dimension sixteen. A set of seven classes involves six nonzero rows, and $\binom{79}{63}$ compositions are out of reach. Theorem 35 supplies the missing leverage: a dominating set of size $7 < n/2$ has an isolated selected vertex. Translate it to zero. Every other selected class is then nonadjacent to the zero class, so it has a representative of weight 2, 4, or 6. Enumerate four of the six other centers, in nondecreasing weight, up to coordinate permutations: there are 104,742 such sorted matrices among the 18,789,795 compositions with four even rows. For each one, let x_0 be the first class not yet dominated. One of the two remaining centers must dominate x_0 and hence lies in its closed neighborhood, which has $1 + \frac{1}{2} \binom{16}{8} = 6436$ members; the last center must then lie in the intersection of the closed neighborhoods of all classes still undominated. The program examines all 385,260,636 admissible choices of the sixth center and finds that this intersection never contains an admissible class. In fact at least 564 classes remain undominated before the last center is placed. Hence no dominating set of size seven exists.

A second, independently written program uses a different decomposition. The 120 classes of weight two can be dominated only by a selected class of representative weight six, which dominates the $\binom{10}{2} = 45$ pairs disjoint from it, or by a selected weight-two class, which dominates only itself. Since $2 \cdot 45 + 4 < 120$, at least three of the six other centers have weight six. The program enumerates by stars and bars the 74,959 column-sorted matrices whose first three rows have weight six and whose fourth row has weight 2, 4, or 6, builds neighborhoods from literal Hamming distances with a different choice of class representatives, and branches on the *last* undominated class through a generic recursion. Its 280,418,188 branch nodes yield no completion.

Structure. For $n = 8$ the four classes $0, e_1 \oplus e_2, e_1 \oplus e_3, e_2 \oplus e_3$ of Theorem 38 are pairwise at distance two and dominate Ω_8 . For $n = 12$ the isolate-anchored enumeration with six centers finds no dominating set containing an isolated selected vertex, so every dominating set of size six is total. $\square$

These finite lower bounds are reproducible computational proofs, not Lean kernel certificates. The three enumeration programs (`folded_cube_orbits`, `folded_cube_anchored`, and `folded_cube_pair_audit`, in C++ with Python wrappers) use exact integers and no optimization solver; both sixteen-dimensional programs include controls in dimensions eight and twelve. The dimension-uniform bounds of Theorem 26 and the prefix construction have separate formal proofs. The value 2054 corrects the numerical value 2064 reported after Theorem 3.1 of Gispert-Fernández et al. [13]; the reported value 132 in dimension eight is confirmed.

Conjecture 40. *For every $n \geq 8$ divisible by four, $\gamma(\Omega_n) = n/2$; equivalently $\xi(Q_n) = 2^{n-1} + n/2$.*

By Theorem 35 and Proposition 37, the conjecture is equivalent to the following statement about first-moment dipole fields: for $n \geq 8$, no product of $n/2 - 1$ linear forms $\langle \sigma(b), y \rangle$ with b

n	vertices of Ω_n	degree	$n/2 - 1$	$\gamma(\Omega_n)$	$\xi(Q_n)$
4	4	3	1	1	9
8	64	35	3	4	132
12	1024	462	5	6	2054
16	16384	6435	7	8	32776
20	262144	92378	9	9 or 10	524297 or 524298

Table 2: The folded even orthogonality graph and the equidistant dimension. Only $n = 4$ attains the lower bound of Theorem 35 among the dimensions decided so far.

even vanishes at every even sign vector other than $\pm\sigma(b)$ for some of its own factors. The quantitative bound (16) guarantees only one unbalanced class in this range, whereas the computed minima of undominated classes are 4, 20, and at least 564 before the final center in dimension sixteen; closing that gap requires information beyond the degree of f_B .

Proof and formalization scope. The lower bound $\xi(Q_{4q}) \geq 2^{4q-1} + 2q - 1$ is a Lean theorem for actual hypercube graph distance, proved through parity orthogonality for products of fewer than n linear forms, and it is combined formally with the prefix equalizer. The three-word common-middle-point theorem and the four-center obstruction are also formal. The general polynomial statement of Theorem 34, the quotient graph as a separate object, Proposition 37, and the finite values above are paper proofs or exhaustive computations, as indicated.

8 Boundary parity on even bishop boards

The remaining sections apply the certificate method to domination problems outside tomography. The interval-cut proof uses boundary incidences to turn local constraints into a global obstruction, and the corner-free lower bound of Theorem 14 was obtained by analysing equality in a counting bound. The same kind of equality analysis settles a recent chessboard conjecture in the positive direction.

Let B_n have vertex set $\{0, \dots, n-1\}^2$, with two distinct squares adjacent when they lie on a common diagonal. A 2-dominating set S requires at least two neighbors in S for every vertex outside S ; vertices in S have no domination requirement. This is ordinary 2-domination, not double total domination. Burchett [6, Conjecture 5.1] conjectured that its minimum cardinality is $2n$ for every even n .

The upper bound is already supplied by [6, Theorem 2.12]. We prove the matching lower bound by excluding equality in the boundary count, and give a boundary-column realization of the known upper bound. The source instead uses the two center columns.

Theorem 41 (Even bishop boards). *For every positive even integer n , $\gamma_2(B_n) = 2n$.*

Proof. The two colors are disconnected and, for even n , are isomorphic by reflection. It suffices to show that the even color requires at least n bishops. Let S be a 2-dominating set within that color, and let ∂E be its boundary squares. Then $|\partial E| = 2(n-1)$.

For a boundary square z , define its weighted coverage

$$c(z) = 2\mathbf{1}_S(z) + |N(z) \cap S|.$$

Every $c(z) \geq 2$. Conversely, each interior bishop attacks at most four boundary squares, and each noncorner boundary bishop contributes its own weight two and attacks at most two other boundary squares. A corner bishop contributes at most three in total. Thus

$$4(n-1) \leq \sum_{z \in \partial E} c(z) \leq 4|S|.$$

In particular $|S| \geq n - 1$. Suppose equality holds. Every boundary square then has coverage exactly two, and no corner belongs to S .

Let p_s be the number of selected bishops on the diagonal $x + y = s$, and r_d the number on $x - y = d$. At every boundary square (x, y) ,

$$c(x, y) = p_{x+y} + r_{x-y} = 2.$$

Indeed, the two diagonal populations count the square twice if it is selected, and otherwise count exactly its selected neighbors. For every even s with $0 \leq s \leq n - 1$, apply this identity at $(s, 0)$ and $(n - 1, n - 1 - s)$. Their difference-diagonal index is the same, so

$$p_s = p_{2n-2-s}.$$

This identity also holds for the larger even indices by reflection. The involution $s \mapsto 2n - 2 - s$ has no fixed even index, since its only fixed index is $n - 1$, which is odd. Hence

$$|S| = \sum_{s \text{ even}} p_s$$

is even, contradicting $|S| = n - 1$. Therefore at least n bishops are needed on each color.

For the upper bound, occupy both outermost columns of the board. There are $2n$ selected squares. An unselected square (x, y) has $1 \leq x \leq n - 2$. Each of its two diagonals meets at least one occupied column: otherwise, for the rising diagonal one would have both $y < x$ and $y > x$, and for the falling diagonal both $x + y < n - 1$ and $x + y > n - 1$. The two diagonal intersections are distinct because their only common square is the unselected square itself. Thus every unselected square has at least two selected neighbors. When $n = 2$, all squares are occupied. This proves the upper bound and the theorem. $\square$

The obstruction is an equality constraint, rather than a stronger average-degree inequality. Boundary counting first forces a candidate optimum into an affine system of diagonal populations. Reflection pairs those populations, and their parity excludes equality. This gives a template for other line-of-sight graphs: retain the equality conditions of a covering bound and look for an involution without fixed lines.

9 Modular certificates: two Petersen domination formulas

The generalized Petersen graph $P(n, k)$ has outer vertices u_i and inner vertices v_i , indexed modulo n , with edges

$$u_i u_{i+1}, \quad u_i v_i, \quad v_i v_{i+k}, \quad n \geq 2k + 1, \quad k \geq 1.$$

This exchanges the letters used for the two layers in [34], without changing the graph. A double total dominating set is a vertex set D such that every vertex has at least two neighbors in D ; its minimum size is $\gamma_{\times 2, t}(G)$.¹

Conjecture 3.1 of Zhao and Wei [34, p. 912] predicts, for positive m , the value $\lceil 4n/3 \rceil$ for $P(n, 3m + 1)$ when $n \not\equiv 2 \pmod{6}$, and one more when $n \equiv 2 \pmod{6}$. Their Conjecture 3.2 predicts $\lceil 4n/3 \rceil$ for $P(n, 3m + 2)$ when $n \equiv 0, 1 \pmod{3}$, and one more otherwise. Thus they respectively predict 15 at $(n, k) = (11, 4)$ and 22 at $(16, 5)$.

Theorem 42.

$$\gamma_{\times 2, t}(P(11, 4)) = 16, \quad \gamma_{\times 2, t}(P(16, 5)) = 24.$$

Consequently both conjectured formulas are false.

¹The introductory definitions on p. 906 of [34] print $\leq k$, but the abstract on p. 905 and the proof of Theorem 2.1 use the required lower bound $\geq k$. We follow that consistent intended convention.

Our purpose is to give explicit, independently checkable refutations with compact certificates. We do not assert that these are the first published counterexamples. An older paper by Sun and Shao [32] studies the same domination parameters; its exact parameter scope could not be inspected in the public sources available for this preparation. The primary conjecture statements and the contradictions proved here are unambiguous, independently of that priority issue. The graph isomorphisms $P(11, 4) \cong P(11, 3)$ and $P(16, 5) \cong P(16, 3)$ were included in the priority search: $4 \cdot 3 \equiv 1 \pmod{11}$ and $5 \cdot 3 \equiv -1 \pmod{16}$. No accessible source inspected in that search established these exact double-total-domination values.

9.1 The transformation: domination, packing, and defects

An open packing is a set C whose open neighborhoods are pairwise disjoint [18]. Write $\rho^o(G)$ for the maximum size of such a set. In a cubic graph,

$$D \text{ is double total dominating} \iff V(G) \setminus D \text{ is an open packing.} \quad (18)$$

Indeed, at least two neighbors in D is equivalent to at most one neighbor in its complement. Hence

$$\gamma_{\times 2, t}(G) = |V(G)| - \rho^o(G).$$

This elementary complement relation is the useful change of problem: lower bounds for domination become obstructions to almost-complete disjoint coverings by neighborhoods.

Let A be a zero-one incidence matrix, whose columns are sets to be packed. If c is the indicator of a packing, then Ac has entries zero or one. Its uncovered defect is

$$d = \mathbf{1} - Ac.$$

For any matrix W over a field with $WA = 0$ and $W\mathbf{1} = 0$, the defect necessarily satisfies $Wd = 0$. In particular, a one-point defect is impossible if every column of W is nonzero. The following version makes the certificates explicit.

Lemma 43 (Modular single-defect obstruction). *Let X be a finite ground set and $\mathcal{B}$ a family of its subsets. Suppose functions $w_1, \dots, w_t : X \rightarrow F$, with values in a field, satisfy*

$$\sum_{x \in X} w_j(x) = 0, \quad \sum_{x \in B} w_j(x) = 0 \quad (B \in \mathcal{B}),$$

for every j , and for every $x \in X$ at least one $w_j(x)$ is nonzero. No pairwise disjoint subfamily of $\mathcal{B}$ covers exactly $|X| - 1$ points.

Proof. If the uncovered point were x , the sum of each weight on the covered points would be zero. Subtracting from its zero total gives $w_j(x) = 0$ for every j , contrary to the hypothesis. $\square$

For $d > 0$, if such weights exist for the open neighborhoods of a d -regular graph of order $dq + 1$, the lemma rules out an open packing of size q . Counting already rules out larger packings, so $\rho^o(G) \leq q - 1$. The same argument applies separately to the two sides of a regular bipartite graph. The certificate is elementary linear algebra; the application lies in finding useful small-field vectors for the particular graph and defect size.

9.2 A modulus-23 certificate for $P(11, 4)$

In outer and inner order, assign the following weights in $\mathbb{F}_{23}$:

$$\begin{aligned} w_u &= (1, 3, 17, 7, 2, 5, 5, 2, 7, 17, 3), \\ w_v &= (17, 5, 13, 4, 11, 16, 16, 11, 4, 13, 5). \end{aligned}$$

All coordinates are nonzero. For each index i modulo 11, direct addition gives

$$w_u(i-1) + w_u(i+1) + w_v(i) = 0, \quad w_v(i-4) + w_v(i+4) + w_u(i) = 0. \quad (19)$$

The total weight is zero. Thus every open neighborhood and the entire graph have zero weight. Seven pairwise disjoint neighborhoods would cover 21 of the 22 vertices, contrary to Lemma 43. Since every neighborhood has size three, $\rho^o(P(11,4)) \leq 6$.

For an integer version of the certificate, take

$$\begin{aligned} \tilde{w}_u &= (-22, 26, 17, -16, 2, 5, 5, 2, -16, 17, 26), \\ \tilde{w}_v &= (17, 5, -10, -19, 11, -7, -7, 11, -19, -10, 5). \end{aligned}$$

The adjacency product is $A\tilde{w} = 69e_{u_0}$, and $\sum \tilde{w} = 23$. Reduction modulo 23 gives exactly the displayed certificate.

The packing

$$C = \{u_0, u_5, u_8, v_2, v_3, v_8\}$$

has the following disjoint neighborhoods:

$$\begin{aligned} &\{u_1, u_{10}, v_0\}, \quad \{u_4, u_6, v_5\}, \quad \{u_7, u_9, v_8\}, \\ &\{u_2, v_6, v_9\}, \quad \{u_3, v_7, v_{10}\}, \quad \{u_8, v_1, v_4\}. \end{aligned}$$

Therefore $\rho^o(P(11,4)) = 6$, and (18) yields the first value in Theorem 42.

9.3 Two modulus-3 checks for each side of $P(16,5)$

The graph is bipartite, with parts

$$P = \{u_i : i \text{ even}\} \cup \{v_i : i \text{ odd}\}, \quad Q = \{u_i : i \text{ odd}\} \cup \{v_i : i \text{ even}\}.$$

Each has size 16. A packing contains at most five vertices in each part by neighborhood counting. We exclude five.

Define a weight a , supported on Q , by the following values in $\mathbb{F}_3$:

$$\begin{aligned} a_u &= (0, 1, 0, 2, 0, 2, 0, 1, 0, 1, 0, 2, 0, 2, 0, 1), \\ a_v &= (1, 0, 0, 0, 2, 0, 0, 0, 1, 0, 0, 0, 2, 0, 0, 0). \end{aligned}$$

Its total weight is zero, and every neighborhood has weight zero. These statements can be checked by the two recurrences

$$a_u(i-1) + a_u(i+1) + a_v(i) = 0, \quad a_v(i-5) + a_v(i+5) + a_u(i) = 0 \quad \text{in } \mathbb{F}_3.$$

On Q , its zero coordinates are exactly v_2, v_6, v_{10}, v_{14} . Let b be the rotation by two indices: $b(u_i) = a(u_{i-2})$ and $b(v_i) = a(v_{i-2})$. Its zeros on Q are v_0, v_4, v_8, v_{12} . Thus a, b are jointly nonzero at every point of Q .

Five packing vertices in P would have disjoint neighborhoods covering 15 points of Q . Apply Lemma 43 with ground set Q and weights $a|_Q, b|_Q$ to exclude this. Rotation by one index exchanges P and Q and supplies the same obstruction to five packing vertices in Q . Consequently every open packing has size at most eight.

The bound is attained by

$$C = \{u_i : i \equiv 0, 1 \pmod{4}\}.$$

Its eight spokes have distinct inner endpoints. Its outer-neighbor pairs are also pairwise disjoint: the pattern 1100 on the outer cycle contributes precisely one selected neighbor at every outer vertex. Hence $\rho^o(P(16,5)) = 8$, and $\gamma_{\times 2,t}(P(16,5)) = 32 - 8 = 24$.

The use of two checks is substantive. One zero-sum check need not have full support, but two complementary supports can exclude every one-point defect. The certificate method is naturally a family of parity checks, rather than a search for one universally nonzero vector.

9.4 Residual syndromes beyond one-point defects

The preceding certificates extend to arbitrary rectangular incidence systems and defects of any size. Let A have row set R and column set C , with its columns indicating subsets to be packed. A selected packing has indicator c and uncovered-row set D . Over any semiring,

$$Ac + \mathbf{1}_D = \mathbf{1}_R.$$

For a rectangular matrix W satisfying $WA = 0$, define the support syndrome

$$\sigma_W(D) = \sum_{r \in D} W_{\bullet r}.$$

Theorem 44 (Multi-defect certificate). *Every packing satisfies $\sigma_W(D) = W\mathbf{1}_R$. If no subset of size below ℓ has this syndrome, then every packing with constant column size k satisfies*

$$k|c| + \ell \leq |R|.$$

More generally, exclusion of all supports of exactly size t excludes every packing size a with $ka + t = |R|$.

Proof. Multiplying the residual identity by W gives the syndrome identity. Disjointness and regular column size give $k|c| + |D| = |R|$. The two conclusions follow immediately. $\square$

The reusable operation is to certify the small residual instead of enumerating the large selected set. For a finite field one can compute a left-kernel basis, form its column vectors, and search their small subset sums. The zero target excludes a one-point defect exactly when every column is nonzero; a two-point defect requires two distinct columns summing to zero. Multiple checks are therefore naturally kept together. Rectangular operators permit separate vertex classes, selected radius shells, or restricted sets of admissible neighborhoods. Neither a square matrix nor an invertibility hypothesis is needed. These are elementary linear-algebraic incidence conservation laws, supplied here as a general verified interface for future searches; the general identity itself is not a novelty claim.

For constant positive column size k , only residual cardinalities congruent to $|R|$ modulo k can occur. The accompanying search program starts with the counting bound and tests precisely those candidate residual sizes, comparing every support syndrome with $W\mathbf{1}_R$, which need not be zero. It improves a bound only after a complete exclusion search at the relevant cardinality.

10 A fractional-face conjecture on six vertices

An incidence formulation also makes it easy to distinguish two different types of forced-zero coordinates. For a graph G , let $A = I + A_G$ be its closed-neighborhood matrix. Define

$$\mathcal{D}_G = \operatorname{argmin}_{f \in [0,1]^V, Af \geq \mathbf{1}} \mathbf{1}^T f, \quad \mathcal{P}_G = \operatorname{argmax}_{f \in [0,1]^V, Af \leq \mathbf{1}} \mathbf{1}^T f.$$

A graph is Class I when the two optimal faces intersect and neither is contained in the other. A vertex is domination-null or packing-null when its coordinate is zero throughout $\mathcal{D}_G$ or $\mathcal{P}_G$, respectively. Rubalcaba's dissertation [29, Conjecture 5.2.5, p. 83] conjectures that every Class I graph has a vertex null in both senses. This is a dissertation conjecture; it is not explicitly restated in the later journal article [28].

Theorem 45. *The connected graph on $\{0, 1, 2, 3, 4, 5\}$ with edge set*

$$\{04, 05, 13, 14, 23, 24, 34\}$$

is Class I and has no vertex that is both domination-null and packing-null.

Proof. The closed neighborhoods $N[3] = \{1, 2, 3, 4\}$ and $N[5] = \{0, 5\}$ partition the vertex set. Summing their inequalities gives $\mathbf{1}^T f \geq 2$ for every feasible dominating function and $\mathbf{1}^T f \leq 2$ for every feasible packing function. The following indicator functions all have weight two:

$$\begin{aligned} e &= \mathbf{1}_{\{3,5\}} && \in \mathcal{D}_G \cap \mathcal{P}_G, \\ d &= \mathbf{1}_{\{0,4\}} && \in \mathcal{D}_G \setminus \mathcal{P}_G, \\ p &= \mathbf{1}_{\{1,5\}} && \in \mathcal{P}_G \setminus \mathcal{D}_G, \\ q &= \mathbf{1}_{\{2,5\}} && \in \mathcal{P}_G. \end{aligned}$$

Their stated feasibility follows by inspecting the six closed neighborhoods. For the strict exclusions, d has load two at vertex 4, while p has load zero at vertex 2. Their supports together cover all vertices. Every coordinate is therefore positive in at least one of the two optimal faces, excluding a jointly null vertex. The edge list describes a diamond with a two-edge pendant path, so the graph is connected. $\square$

This proof concerns the full real-valued optimal faces, even though integral vectors suffice as certificates. It does not replace fractional optimization by integer optimization. Non-containment of the two faces can force different zero-coordinate phenomena without forcing their intersection at one vertex. No earlier refutation was located in the inspected follow-up literature; that search is not a guarantee of priority.

The connectedness of this certificate is useful context. Without a connectedness restriction, $P_3 \sqcup P_4$ already yields an elementary counterexample by taking products of optimal faces, as one can infer from the path and disjoint-union classifications in [29, pp. 44 and 51]. On P_3 , domination has the unique optimum $(0, 1, 0)$, while packing also admits $(1/3, 1/3, 1/3)$. On P_4 , packing has the unique optimum $(1, 0, 0, 1)$, while domination also admits $(1/2, 1/2, 1/2, 1/2)$. Their product faces intersect, neither contains the other, and their supports jointly cover every vertex. This is a direct inference, not a located prior publication of a refutation. It rules out using the age of the conjecture as evidence of difficulty. A subscription glossary was not available for inspection, so first-refutation priority remains unconfirmed.

11 Distributed sensing interpretation

The grid theorem applies to a distributed observation contract in which a known grid network carries at most two anonymous events or agents, each labelled monitor reports the complete hop-distance histogram, and one monitor report can be lost. It supplies exact placement guarantees under that contract, including coincident agents. It does not distinguish agent trajectories or tolerate an unlabelled mixture of monitors.

The observation contract can be realized by exact aggregation. For each monitor s , build a rooted breadth-first spanning tree. A vertex v contributes the monomial $x(v)T^{\text{dist}(s,v)}$ and sends its sum with the polynomials received from its children toward the root. Induction on subtrees shows that the root receives the shell polynomial (1), whose coefficients are precisely $H_s x$. Messages carry integer counts indexed by distance; they need not contain target identities. This establishes the report semantics under reliable aggregation. The theorem then tolerates a missing monitor output, not a failure or corruption inside the aggregation tree.

Corollary 9 reduces the final output at each noncorner monitor to the integer $M_1 x(s)$, between 0 and $4(n-1)$ in the stated mass range. Such a monitor can aggregate these scalar contributions directly along its tree. This gives an exact output-compression guarantee, without assumptions about a particular networking protocol.

A missing report and a false report are different faults. By Proposition 5, the constructions have minimum block distance two and therefore detect one corrupt block, but unique correction of one arbitrary corrupt block requires distance at least three. No Byzantine correction claim

is inferred from the erasure theorem. Bandwidth, packet delay, acquisition cost, and numerical noise are outside this model.

The interval-cut reduction also gives a direct placement checker: enumerate equal-length interval pairs and count crossing sensors. There are $\sum_{k=1}^m (m - k + 1)^2 = O(m^3)$ pairs. With a two-dimensional prefix count table, each cut count is obtained in constant time from row, column, and rectangle counts. Thus corner-augmented placements can be checked in $O(m^3 + n^2)$ time, after reading the placement. This algorithm follows the characterization, not enumeration of all populations.

12 Verification

The accompanying package uses Lean 4.35.0-rc2 with the matching pinned Mathlib revision. Foundational population definitions and the square-grid graph-distance adapter originate in the preceding formalization, with pinned dependencies and verification input hashes included in the package.

Formal endpoints. The following statements are Lean theorems about the concrete graphs and observation models of this paper, with actual shortest-path distances.

- *Grids.* The interval-cut criterion (Theorem 8) and the compressed-report equivalence (Corollary 9); the charging bound (6); the leaf-star interval cuts (Lemma 11); and the exact corner-constrained optimum $\lceil (3m+9)/2 \rceil$ for every interior size $m \geq 1$. The ordered-star variant, the strict bound $3m < 2|S|$ without corners, the rectangular criterion, and the long-rectangle optimum with its rigidity are also formal.
- *Hypercubes.* The prefix equalizer and its cardinality; the reduction of equalization to the antipodal cover condition; parity orthogonality for products of fewer than n linear forms; and the lower bound $|S| \geq 2^{4q-1} + 2q - 1$ for every distance equalizer of Q_{4q} , combined with the upper witness into a single two-sided statement. The three-word common-middle-point theorem and the four-center obstruction are formal.
- *Certificates.* The bishop theorem for all even boards, both exact Petersen values, the full real-valued fractional-face counterexample, and the rectangular incidence-syndrome interface.
- *Sensing algebra.* Additive-field connectivity and erasure thresholds, tensor-moment identities, and polynomial interpolation on a lower Hamming ball.

All proved endpoints use only the standard logical axioms; no admitted proof, native-decision axiom, or custom mathematical axiom is accepted. An independent Challenge/Solution interface selects thirty-three principal statements, including the all-order grid optimum and both hypercube bounds, for comparison through the Lean kernel, NanoDa, and con-ron. The Challenge intentionally contains statement holes, whereas the Solution imports only proved development. The recorded local comparison passed; it is not service-side verification, registration, or acceptance.

Paper proofs without a formal endpoint. The quotient graph Ω_n and its domination number as separate objects, the general polynomial form of Theorem 34, Proposition 37, the unrestricted grid lower bound of Theorem 14, the single-moment capacity theorem with its spectral decomposition and sampling consequence, the sharp cube moment-code distance, the hybrid projection capacities, and the cycle-consistency statements for additive fields have complete proofs in this paper and are not Lean endpoints. The distributed aggregation description and the placement-checker complexity are paper-level consequences, not verified protocol implementations.

Exhaustive computations. The values $\gamma(\Omega_8) = 4$, $\gamma(\Omega_{12}) = 6$, and $\gamma(\Omega_{16}) = 8$ of Theorem 39 combine formal upper witnesses with exhaustive lower-bound enumerations, each performed by two independently written programs with complete case counts; they are not presented as kernel-checked. Proposition 15 rests on explicit placements for $3 \leq n \leq 20$ that are re-verified from literal distance bags, with optimality from the proved lower bound. Independent exact programs also inspect literal distance bags and interval cuts, the leaf-star placements for all $m \leq 40$, compressed reports, all 2,083,861 pairs outside the explicit twelve-dimensional equalizer, small cases of the quantitative balancing bound, bishop constructions, Petersen packings, continuous fractional-face certificates, modular residual searches, moment spectra, product witnesses, and rectangular designs. These tests supplement the infinite proofs. No stated result relies on a floating-point optimizer: the integer-programming search used to locate corner-free placements is a discovery aid whose outputs are checked independently.

13 Conclusion and open problems

Distance-shell tomography makes anonymous ambiguity an exact mathematical object. A sensor report is a linear observation of an integer population; two populations are indistinguishable precisely when their difference is a trade; and each sensor either detects a trade or is blind to it. This formulation preserves multiplicity, permits coincident targets, and separates the information available to a monitor from assumptions about target identity. Instead of matching reported targets across sensors, one characterizes the trades and designs sensors against their detector sets [30].

The results show what the calculus achieves beyond a change of notation. On grids, corner reports reduce every remaining ambiguity to interval detectors, and that single family yields a universal lower bound, an exact optimum for every board size, a polynomial-time placement checker, and a compression guarantee. The equality case of the lower bound dictates the optimal design: three-leaf stars, half-lines, and a parity argument. On Hamming products, a single raw moment encodes every low-order marginal, minimal integer trades identify the exact point at which recovery fails, and a classical minimum-support eigenfunction makes the fault threshold exact [33]. On hypercubes, the blind sets of antipodal dipoles are hyperplanes, their first-moment fields multiply to a low-degree polynomial, and a dimension count on the unbalanced points pins the equidistant dimension to two consecutive integers.

The wider applications show how the search for structured residual obstructions guides other graph problems. On bishop boards, equality in a boundary count forces diagonal populations into a parity contradiction. In incidence packings, a large candidate solution is excluded by certifying its small uncovered residual. These are extensions of the same style of reasoning, not automatic consequences of the tomography theorems: each requires its own graph-specific reduction.

Several concrete problems remain.

1. **Close the last unit for hypercubes.** Conjecture 40 asserts $\gamma(\Omega_n) = n/2$ for all $n \geq 8$; the present work leaves it unresolved for $n \geq 20$. By Theorem 35 a counterexample is a dominating set of size $n/2 - 1$ with an isolated member, and by Proposition 37 its dipole product is a fixed combination of truncated point masses. The degree argument alone cannot exclude this, since it allows a single unbalanced class; the arithmetic of products of sign-vector linear forms must enter. A short certificate for the finite cases would also replace the exhaustive computations.
2. **Determine the corner-free grid budget.** Conjecture 16 is verified for $n \leq 20$. A proof needs a criterion replacing Theorem 8 when corners are absent, that is, a classification of the boundary families of two-target trades. For several erasures, corner-invisible

trades require higher interval-cut multiplicity, but corner-visible pairs also need stronger separation because four corners alone supply only two detectors.

3. **Optimize higher-mass and mixed sensing.** The Hamming-product capacity threshold and the sharp three- and four-direction grid contracts give starting points beyond two targets. Determine the minimum number of scalar moment sensors when the population mass is bounded: the field-reconstruction sampling theorem solves the unrestricted field problem, which can require more sensors. For robust cube moments, determine the exact distance below mass 2^{k-1} and in dimensions $d < 2k$, where the disjoint-pair extremizer does not apply.
4. **Extend quantitative balancing.** Theorem 34 is tight for at most one center and far from the computed minima for $n/2 - 1$ centers. Determine the true minimum number of unbalanced classes for each number of centers, and the analogous statement for the relaxed balancing problems of [2].
5. **Automate certificate discovery and verification.** Extend the rectangular syndrome interface to searches over several fields and residual cardinalities, with search outputs translated into Lean-checkable certificates. The search should expose a short obstruction whose validity is independent of the optimizer or enumeration strategy.
6. **Develop distributed sensing under richer contracts.** Study communication cost, monitor placement under geometric constraints, dynamic populations, and partial loss within aggregation. Low-order moments can be accumulated by binomial updates as a message crosses a tree edge; designing economical acquisition protocols is a separate task from proving that the resulting final fields identify populations.

These directions retain the central advantage of distance-shell tomography: an explicit chain from observation semantics to algebraic ambiguity, geometric structure, and checkable design guarantees.

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Availability. This paper and its supporting material are archived on Zenodo: <https://doi.org/10.5281/zenodo.23125523>. The archive includes the editable manuscript and figures, the pinned Lean project, exact verification programs and data, and records of the formal checks and executed verification. The deposit is a preprint; journal acceptance and completed independent human peer review are not claimed.

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